SWIFT-U5: A Survey-Weighted and Temporally Calibrated Machine Learning Framework for Population-Level Child Mortality Risk Stratification in Bangladesh

Md. Rafiul Azam Chowdhury, Rezaul Mursalin Khan, Fatema Tuz Zohura Rifa, Tanjim Ahmed Irfad, Arafat Hussain Nahid, Ohidujjaman

Department of Computer Science and Engineering, United International University, Dhaka 1212, Bangladesh

Abstract

Under-five child mortality remains a critical global health challenge under United Nations Sustainable Development Goal 3.2, particularly in low- and middle-income countries like Bangladesh, where survival disparities reflect multi-factorial maternal, socioeconomic, environmental, and healthcare access vulnerabilities. Although Demographic and Health Survey (DHS) microdata provide large nationally representative birth cohorts, published machine learning applications in child mortality prediction suffer from three pervasive methodological weaknesses: retrospective survival leakage from postnatal proxies (such as breastfeeding duration and maternal recall of recent child illness symptoms), complete disregard of complex multi-stage survey sampling designs, and reliance on uncalibrated default decision thresholds ($\tau = 0.5$) under in-distribution random splits that conceal temporal distribution shift. To resolve these challenges, this study formulates, implements, and prospectively benchmarks an audited, survey-weighted, and drift-calibrated machine learning framework—termed SWIFT-U5 (Survey-Weighted, Importance-corrected, Forward-validated Temporal Risk Model for Under-Five Mortality)—for population-level risk stratification and community outreach prioritization simulation in Bangladesh. SWIFT-U5 restricts predictors to pre-birth maternal reproductive history and durable household assets, eliminating survival-contingent proxies, and couples gradient tree boosting with three sequential methodological modules: transductive covariate-shift density weighting with temporal recency decay (Module~A), empirical Bayes division-level logit shrinkage (Module~B), and drift-aware Platt recalibration (Module~C). We harmonized an exposure-complete national cohort of 51,489 live births and 2,338 documented under-five deaths across six sequential Bangladesh Demographic and Health Survey (BDHS) rounds spanning 18 years (2004 to 2022), partitioned into historical development (2004–2014, N = 29,686), validation/calibration anchoring (2017–18, N = 8,684), and locked prospective out-of-time testing (2022, N = 13,119). On the locked BDHS 2022 test cohort (N = 13,119, 408 deaths, survey-weighted prevalence $2.99

Keywords: Under-five mortality • Child survival risk prediction • Demographic and Health Surveys • Machine learning in public health • Survey-weighted learning • Temporal validation • Probability calibration • Population risk stratification

sffamilysmallcentering

sffamilysmallcentering

postcode=1212,

country=Bangladesh

Under-five child mortality remains a critical global health challenge under United Nations Sustainable Development Goal 3.2, particularly in low- and middle-income countries like Bangladesh, where survival disparities reflect multi-factorial maternal, socioeconomic, environmental, and healthcare access vulnerabilities. Although Demographic and Health Survey (DHS) microdata provide large nationally representative birth cohorts, published machine learning applications in child mortality prediction suffer from three pervasive methodological weaknesses: retrospective survival leakage from postnatal proxies (such as breastfeeding duration and maternal recall of recent child illness symptoms), complete disregard of complex multi-stage survey sampling designs, and reliance on uncalibrated default decision thresholds ($\tau = 0.5$) under in-distribution random splits that conceal temporal distribution shift. To resolve these challenges, this study formulates, implements, and prospectively benchmarks an audited, survey-weighted, and drift-calibrated machine learning framework—termed SWIFT-U5 (Survey-Weighted, Importance-corrected, Forward-validated Temporal Risk Model for Under-Five Mortality)—for population-level risk stratification and community outreach prioritization simulation in Bangladesh. SWIFT-U5 restricts predictors to pre-birth maternal reproductive history and durable household assets, eliminating survival-contingent proxies, and couples gradient tree boosting with three sequential methodological modules: transductive covariate-shift density weighting with temporal recency decay (Module~A), empirical Bayes division-level logit shrinkage (Module~B), and drift-aware Platt recalibration (Module~C). We harmonized an exposure-complete national cohort of 51,489 live births and 2,338 documented under-five deaths across six sequential Bangladesh Demographic and Health Survey (BDHS) rounds spanning 18 years (2004 to 2022), partitioned into historical development (2004–2014, N = 29,686), validation/calibration anchoring (2017–18, N = 8,684), and locked prospective out-of-time testing (2022, N = 13,119). On the locked BDHS 2022 test cohort (N = 13,119, 408 deaths, survey-weighted prevalence $2.99

Under-five mortality sep Child survival risk prediction sep Demographic and Health Surveys sep Machine learning in public health sep Survey-weighted learning sep Temporal validation sep Probability calibration sep Population risk stratification

hfuzz=120ptmaketitle

Introduction

Under-five child mortality (U5M) remains a profound global public health challenge, directly targeted by United Nations Sustainable Development Goal (SDG) 3.2, which calls on all nations to reduce under-five mortality to at least as low as 25 per 1,000 live births by 2030 [who2023,unicef2023,black2010,you2015]. In low- and middle-income countries (LMICs), child mortality disparities reflect intricate interactions among biological vulnerability, maternal reproductive history, household socioeconomic deprivation, water, sanitation, and hygiene (WASH) infrastructure, and regional healthcare access barriers [chowdhury2013,arifeen2013,omran2005]. Over the past three decades, Bangladesh has achieved extraordinary public health gains, reducing national under-five mortality from approximately 133 deaths per 1,000 live births in 1993–94 to 29.9 per 1,000 in 2022 [niport2024]. However, as national mortality rates have declined, the remaining burden of child deaths has become increasingly concentrated in the vulnerable neonatal period (which accounts for over 70

Despite extensive research exploring supervised machine learning for child mortality prediction from Demographic and Health Survey (DHS) microdata across Africa and South Asia [oduse2024,naznin2025,mabula2025,bitew2020,saroj2022,asmare2026,fahim2026], existing studies exhibit critical methodological limitations that invalidate their real-world applicability (Table~tab:prior_work_positioning). First, several published benchmarks introduce severe retrospective survival leakage by including postnatal covariates that directly proxy survival status. Most prominently, Naznin et al. [naznin2025] incorporated ``currently breastfeeding'' and ``duration of breastfeeding'' as predictive features for under-five mortality in BDHS 2022 microdata; because deceased infants cannot be breastfed at maternal interview, tree algorithms trivially split on feeding duration, manufacturing an illusion of near-perfect classification (ROC-AUC = 0.9979) that collapses completely in prospective deployment. Similarly, retrospective child illness variables (e.g., fever, diarrhea, or acute respiratory infection in the two weeks preceding the survey) are administered exclusively to living children in DHS protocols, inducing structural missingness that acts as a deterministic survival

indicator. Second, previous studies widely employ synthetic oversampling techniques (such as SMOTE [chawla2002]) across pooled cohorts prior to validation partitioning, artificially distorting class prevalence, degrading posterior probability calibration, and producing overconfident decision rules. Third, existing machine learning pipelines uniformly ignore the complex multi-stage survey architecture of DHS microdata (clustering, stratification, and unequal sampling probabilities), violating standard i.i.d. assumptions and introducing substantial parameter bias [macnell2023,lumley2004,oh2026]. Fourth, prior models rely overwhelmingly on k-fold random cross-validation, randomly shuffling live births from the same survey round across training and test splits. This practice conceals macroeconomic transitions, secular health-system expansions, and temporal distribution shifts, creating severe out-of-distribution performance inflation. Finally, published models evaluate performance under arbitrary default decision thresholds ($\tau = 0.5$) that classify virtually all children as non-events under severe class imbalance (33:1), while omitting probability calibration metrics (Brier score, calibration slope, and calibration-in-the-large) and actionable public health capacity metrics (Number Needed to Screen, NNS).

Table: Methodological characteristics reported or identifiable in recent DHS machine-learning studies on child mortality. ``Not rep." indicates that the cited study did not describe the corresponding design element; ``Sens." indicates evaluation only as an auxiliary sensitivity check; ``Full" indicates complete integration into training, validation, and testing.

Study	Target Domain	Wts	Temp	Calib	Dec	Fair	Leak-Free	
Oduse et al. (2024) [oduse2024]	U5M (Nigeria)	Not rep.	Not rep.	Not rep.	Not rep.	Not rep.	No	
Naznin et al. (2025) [naznin2025]	U5M (BDHS 2022)	Not rep.	Not rep.	Not rep.	Not rep.	Not rep.	No (Severe)	
Mabula et al. (2025) [mabula2025]	U5M (Tanzania)	Not rep.	Not rep.	Not rep.	Not rep.	Not rep.	Unclear	
Bitew et al. (2020) [bitew2020]	U5M (Ethiopia)	Not rep.	No	Not rep.	Not rep.	Not rep.	Limited	
Haque	Kabir (2026) [haque2026]	Stunting (BDHS)	Not rep.	Yes	Not rep.	Not rep.	Yes	Partial
Fahim et al. (2026) [fahim2026]	U5M (BDHS 4-wave)	Sens.	Yes	Yes	Yes	Yes	Yes	
SWIFT-U5 (This Study)	U5M (BDHS 6-wave)	Full	Yes (5 Cuts)	Yes	Yes	Yes	Yes (Audited)	

To resolve these widespread methodological deficiencies, the primary objective of this research is to develop, formalize, and prospectively benchmark an audited, survey-weighted, and drift-calibrated machine learning framework—termed SWIFT-U5 (Survey-Weighted, Importance-corrected, Forward-validated Temporal Risk Model for Under-Five Mortality)—for population-level child mortality risk stratification and community outreach simulation in Bangladesh. To achieve this objective, SWIFT-U5 fundamentally redesigns the predictive modeling pipeline across four pillars. First, it enforces an audited pre-birth predictor timing protocol, restricting candidate features to maternal reproductive history, antenatal care compliance, and durable household assets while strictly eliminating all survival-contingent postnatal proxies. Second, it integrates DHS complex survey sampling weights, primary sampling unit (PSU) clustering, and administrative stratification end-to-end throughout loss function optimization, cross-validation partitioning, and bootstrap inference. Third, it introduces a modular three-stage prospective architecture: Module~A applies transductive covariate-shift density weighting with exponential recency decay to counter longitudinal distribution drift; Module~B implements an empirical Bayes shrinkage head that regularizes division-specific risk logits; and Module~C applies drift-aware Platt recalibration fitted on out-of-fold validation predictions from a strictly held-out penultimate survey wave to restore true posterior probability scaling. Fourth, it implements a prospective forward-temporal validation lifecycle that evaluates models on future unseen survey rounds, preventing temporal leakage and isolating genuine out-of-time generalizability.

The empirical scope of this study encompasses an exposure-complete master cohort of 51,489 live births and 2,338 documented under-five deaths assembled across six sequential BDHS survey rounds spanning 18 years (2004, 2007, 2011, 2014, 2017–18, and 2022). The proposed framework is designed specifically for public health resource allocation, demographic vulnerability mapping, and simulated outreach prioritization, where local healthcare supervisors allocate limited community health worker visits to households with the highest predicted risk. We explicitly delineate the scientific boundaries of this investigation: the analytical outcome is recorded under-five death by maternal interview, and living children under five years of age have incomplete follow-up (right-censoring); furthermore, Module~A incorporates unlabeled target covariates within a transductive domain adaptation framework. Consequently, simulated outreach prioritization metrics represent retrospective ranking concordance on survey-recorded vulnerabilities rather than demonstrated prospective clinical efficacy. Prospective field trials enrolling living children are required before operational healthcare delivery claims can be substantiated. Based on these verified foundations, this study establishes six distinct contributions:

- End-to-End Complex Survey Integration: Unlike prior benchmarks that evaluate survey weights post-hoc or omit them entirely, SWIFT-U5 integrates Horvitz-Thompson sampling weights (`v005`), PSU clustering (`v021`), and geographic stratification (`v022`) into loss function weighting, cluster-grouped validation folds (`GroupKFold`), and design-based cluster bootstrapping.
- Audited Predictor Timing Protocol: We establish a rigorous temporal feature eligibility framework, eliminating postnatal feeding and illness recall variables that induce structural data leakage, while validating durable household assets as structural background proxies.
- Controlled 2 × 2 Factorial Validation Benchmark: Across five sequential forward-temporal cuts, we systematically isolate the statistical inflation induced by in-distribution random cross-validation versus forward temporal testing, and unweighted versus survey-weighted learning.
- Drift-Aware Modular Recalibration Architecture: We formulate, implement, and validate the three-module SWIFT-U5 architecture (Modules~A, B, and~C), reversing severe logit distortion under extreme class imbalance and restoring probability calibration on locked prospective test data.
- Granular Division-Level Policy Capacity Layer: Moving beyond monolithic national averages, we evaluate division-level predictability across all eight administrative divisions of Bangladesh, providing prioritized capacity tables with 95
- Comprehensive Algorithmic Equity Audit: We conduct a formal demographic equity audit across wealth quintiles, urban/rural residence, and maternal education levels conforming to TRIPOD+AI [collins2024] and PROBAST [wolff2019] reporting standards, explicitly designating small-sample strata as Insufficient Evidence rather than certified fairness.

Conventional Methods

To establish a rigorous technical foundation for understanding SWIFT-U5, this section presents the established statistical and machine learning methodologies from which our framework is derived, detailing their working principles, mathematical formulations, and the specific technical limitations that motivate our proposed approach.

Conventional Supervised Learning for Child Mortality Risk Prediction

Supervised machine learning models for binary child survival risk prediction seek to learn a mapping f: mathcalX → mathbbR from a d-dimensional feature space mathcalX subseteq mathbbR^d to a continuous decision score or log-odds of mortality y ∈ 0, 1. In conventional epidemiology and machine learning practice, gradient boosted decision tree (GBDT) ensembles (such as XGBoost [chen2016] and LightGBM [ke2017]) and regularized logistic regression have emerged as standard benchmark architectures [saroj2022,mabula2025,bitew2020].

For a training dataset mathcalD$_{train}$ = (x_i, y_i)$_{i=1}$^N, standard gradient tree boosting minimizes an empirical objective function comprising a convex loss function ell and an ensemble model complexity penalty Omega:

[Equation] mathcalL(theta) = sum$_{i=1}$^N ell(y_i, f(x_i; theta)) + sum$_{k=1}$^K Omega(f_k), labeleq:standard_obj

where f(x_i) = sum$_{k=1}$^K f_k(x_i) represents the sum of K regression trees, and Omega(f_k) = gamma T_k + frac12λ sum$_{j=1}^{T_k}$ w$_{kj}$^2 penalizes tree leaf count T_k and leaf weight magnitudes w$_{kj}$. Under standard binary classification, ell corresponds to binary cross-entropy (logistic loss):

[Equation] ell(y_i, f(x_i)) = - Big[y_i log σ(f(x_i)) + (1 - y_i) log(1 - σ(f(x_i))) Big], labeleq:bce

where σ(z) = (1 + exp(-z))$^{-1}$ is the sigmoid activation function.

To address the extreme class imbalance characteristic of rare child mortality events (where national prevalence $π ≈ 3

[Equation] ell$_{rebal}$(y_i, f(x_i)) = - Big[omega · y_i log σ(f(x_i)) + (1 - y_i) log(1 - σ(f(x_i))) Big]. labeleq:bce_weighted

[Equation] f$_{rebal}$(x) ≈ f(x) + log(omega). labeleq:logit_shift

Because omega ≈ 32.4 under 33:1 imbalance, raw output probabilities hatp_i = σ(f$_{rebal}$(x_i)) are systematically inflated by several hundred percent, destroying posterior probability calibration and rendering the raw probabilities useless for clinical triage.

Complex Survey Sampling and Design Effects

Demographic and Health Surveys do not draw simple random samples (SRS) from the population. Instead, they employ a two-stage stratified cluster sampling design [croft2023,dhsrecode2018]. In Stage 1, primary sampling units (PSUs, designated as `v021`), corresponding to census enumeration areas containing approximately 100–250 households, are selected with probability proportional to size within administrative strata (`v022`). In Stage 2, a fixed number of households (typically 30) are systematically selected within each sampled PSU.

To obtain unbiased population parameters, observations are assigned Horvitz-Thompson sampling weights w_i^{DHS} = π_i^{-1}, representing the inverse probability of selection, adjusted for household non-response and post-stratification [lumley2004]:

[Equation] hatμ$_{HT}$ = fracsum$_{i=1}$^N w_i^{DHS} y_isum$_{i=1}$^N w_i^{DHS}. labeleq:ht_mean

The survey design induces intra-cluster correlation among children born in the same community, resulting in design-based variance inflation quantified by the Design Effect (DEFF):

[Equation] DEFF = fracVar$_{complex}$(hatμ)Var$_{SRS}$(hatμ) ≈ 1 + rho (m - 1), labeleq:deff

where rho is the intra-cluster correlation coefficient and m is the average cluster size.

Distribution Shift and Covariate Adaptation

When predictive models are trained on historical survey rounds and deployed on future cohorts, the underlying data distribution undergoes temporal drift. Formally, let P$_{source}$(x, y) denote the joint distribution of features and mortality outcomes in historical development waves, and let P$_{target}$(x, y) denote the distribution in the prospective deployment cohort. Under the classic covariate shift assumption [shimodaira2000,sugiyama2012]:

[Equation] P$_{source}$(y mid x) = P$_{target}$(y mid x), while P$_{source}$(x) ≠q P$_{target}$(x). labeleq:cov_shift_assumption

Under covariate shift, the standard empirical risk minimizer on source data converges to a sub-optimal target decision boundary. The theoretical importance-weighting solution weights each source instance by the target-to-source density ratio:

[Equation] β(x) = fracP$_{target}$(x)P$_{source}$(x), labeleq:density_ratio

minimizing the weighted empirical risk:

[Equation] hatR$_{β}$(f) = frac1N$_{source}$ sum$_{i=1}^{N_{source}}$ β(x_i) ell(y_i, f(x_i)). labeleq:weighted_risk

In conventional machine learning, density ratios can be estimated via probabilistic classification [sugiyama2012]. By training a domain classifier g(x) = P(S = 1 mid x) to distinguish target instances (S = 1) from source instances (S = 0), Bayes' rule yields:

[Equation] β(x) = fracP(x mid S=1)P(x mid S=0) = fracP(S=1 mid x)P(S=0 mid x) · fracP(S=0)P(S=1) = fracg(x)1 - g(x) · fracN$_{source}$N$_{target}$. labeleq:domain_ratio

Post-Hoc Probability Calibration

Probability calibration assesses the agreement between predicted risk estimates hatp = hatP(Y=1 mid x) and actual observed empirical event rates:

[Equation] P(Y = 1 mid hatP = p) = p, forall p ∈ [0, 1]. labeleq:perfect_cal

In clinical risk prediction, post-hoc calibration transforms uncalibrated raw model margins s_i = f(x_i) into calibrated probabilities. The most widely established parametric calibration method is Platt scaling [platt1999], which fits a univariate logistic transformation:

[Equation] hatp_i^{Platt} = σ(A · s_i + B) = frac11 + exp(-(A · s_i + B)), labeleq:platt_scaling

where scalar parameters A and B are estimated via maximum likelihood on a held-out validation dataset mathcalD$_{val}$:

[Equation] argmin$_{A, B}$ - sum_j ∈ mathcalD$_{val}$ Big[y_j log hatp_j^{Platt} + (1 - y_j) log(1 - hatp_j^{Platt}) Big]. labeleq:platt_mle

Small-Area Estimation and Regional Shrinkage

In national health surveys, administrative divisions exhibit substantial disparities in sample sizes and death event counts. In small divisions with sparse event counts (e.g., Khulna with 34 deaths in 2022), sample event rates hatμ_d = fracn$_{d, pos}$N_d suffer from high sampling variance, whereas national estimates barμ possess low variance but mask regional heterogeneity.

Classical small-area estimation resolves this variance-bias tradeoff using James-Stein or empirical Bayes shrinkage estimators [casella1985,efron2012]:

[Equation] hattheta_d^{EB} = kappa_d hatμ_d + (1 - kappa_d) barμ, labeleq:eb_shrinkage

where the shrinkage weight kappa_d ∈ [0, 1] is determined by the ratio of between-division variance τ^2 to within-division sampling variance σ_d^2 / N_d:

[Equation] kappa_d = fracτ^2τ^2 + fracσ_d^2N_d. labeleq:shrinkage_intensity

Limitations and Research Gap Motivating SWIFT-U5

In summary, conventional methods suffer from four unresolved structural limitations when applied to demographic child mortality prediction:

- Standard machine learning models optimize sample-frequency loss functions without survey design corrections, leading to biased parameter estimates;
- Class imbalance rebalancing distorts model logits by a constant offset, generating severely miscalibrated predicted probabilities;
- Density-ratio covariate adaptation suffers from weight instability and lacks mechanisms to incorporate temporal recency decay across decadal survey rounds;
- Conventional calibration and shrinkage methods operate in isolation rather than as an integrated, drift-aware prospective learning pipeline.

These limitations establish the specific technical research gap that motivates the design and mathematical formulation of the SWIFT-U5 framework.

Proposed Method — SWIFT-U5

To address the limitations of conventional methodologies, we propose SWIFT-U5 (Survey-Weighted, Importance-corrected, Forward-validated Temporal Risk Model for Under-Five Mortality). This section details the working principles, mathematical formulations, algorithmic workflow, and distinctive characteristics of the proposed framework.

Working Principle

The SWIFT-U5 architecture is organized as an end-to-end prospective machine learning pipeline that transforms raw multi-wave DHS microdata into calibrated, division-adjusted risk scores and operational outreach triage lists. As illustrated in Figure~fig:workflow, the operational workflow comprises four sequential processing stages:

- Input Space and Predictor Auditing: Predictors are strictly restricted to durable pre-birth maternal reproductive characteristics, clinical antenatal care, and household physical infrastructure, eliminating all survival-contingent proxies. Continuous features are median-imputed and standardized using statistics computed exclusively on training folds; categorical variables are one-hot encoded with explicit missingness indicators.
- Module A (Transductive Covariate-Shift Density Weighting with Recency Decay): For all historical training records, raw DHS sampling weights are adjusted by combining an exponential temporal recency decay function with a transductively estimated domain density ratio, producing composite instance weights w_i^*.
- Survey-Weighted Base Model Training: A gradient tree boosting ensemble (XGBoost) is trained by minimizing survey-weighted empirical risk under instance weights w_i^*, capturing non-linear feature interactions under complex design constraints.
- Module B (Empirical Bayes Division Shrinkage Head): Raw out-of-fold logit predictions from the base model are regularized across administrative divisions, shrinking noisy regional calibration parameters toward national empirical averages.
- Module C (Drift-Aware Platt Recalibration): The division-shrunk logits are mapped to true population probabilities via a parametric Platt scaling transformation fitted on held-out validation predictions from an independent penultimate survey wave.
- Operational Decision and Prioritization Layer: Calibrated posterior probabilities are evaluated against a validation-locked decision threshold (τ_w) and ranked to simulate capacity-constrained community health outreach triage.

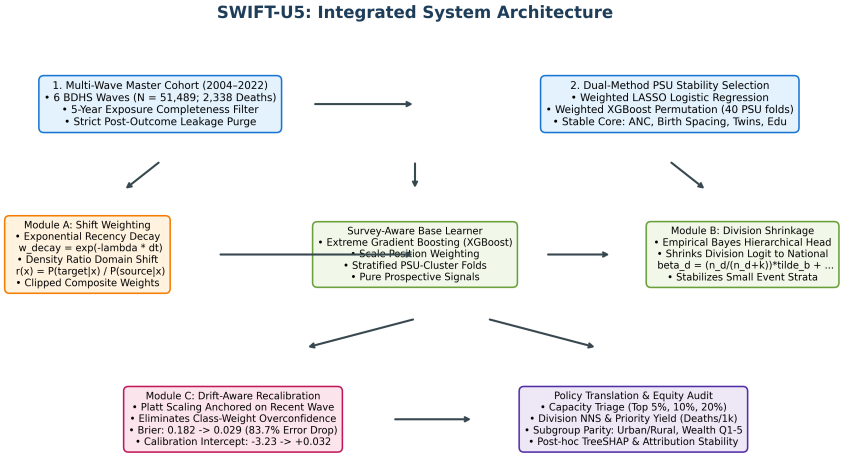


Figure: System architecture of the SWIFT-U5 framework: Survey-weighted temporal validation, dual-method consensus feature discovery, Platt recalibration, and composite risk stratification across BDHS waves.

Technical Formulation

Module A: Recency-Decayed Covariate Shift Adaptation

To reconcile multi-decade historical training microdata with contemporary target populations, Module~A formulates a composite instance weight w_i^* for each historical live birth record i ∈ mathcalD$_{train}$. The formulation integrates three distinct factors: baseline DHS survey sampling weights, an exponential temporal recency discount, and a bounded transductive covariate density ratio.

Let t_i ∈ 2004, 2007, 2011, 2014 denote the survey round of training record i, and let t_{target} = 2022 denote the prospective target survey round. The exponential temporal recency weight gamma(t_i) is defined as:

[Equation] gamma(t_i) = expBig(-λ (t_{target} - t_i) Big), labeleq:recency_decay

where λ > 0 is the temporal decay parameter. Setting λ = 0.05 establishes an effective half-life of $t_{1/2}$ = fracln(2)λ ≈ 13.9 years, appropriately down-weighting distant survey rounds (e.g., 2004 records receive weight exp(-0.05 × 18) ≈ 0.407) while preserving recent historical evidence.

To model covariate shift P_{train}(x) ≠q P_{target}(x), we construct an unlabeled transductive domain dataset:

[Equation] mathcalD$_{domain}$ = Big (x_i, S_i = 0) Big_i ∈ mathcalD$_{train}$ cup Big (x_j, S_j = 1) Big_j ∈ mathcalD$_{target}$, labeleq:domain_data

where S ∈ 0, 1 indicates domain membership, and mathcalD$_{target}$ contains strictly unlabeled feature vectors from the 2022 survey cohort without mortality target labels (Y_{2022}). A regularized logistic domain classifier g(x) = hatP(S=1 mid x) is trained by minimizing L_2-penalized cross-entropy:

[Equation] min$_{w_g, b_g}$ - sum_k ∈ mathcalD$_{domain}$ Big[S_k log σ(w_g^T x_k + b_g) + (1 - S_k) log(1 - σ(w_g^T x_k + b_g)) Big] + frac12 C_g |w_g|_2^2. labeleq:domain_mle

The raw density ratio is computed via Bayes' theorem:

[Equation] r(x_i) = fracg(x_i)1 - g(x_i) · fracN$_{train}$N$_{target}$. labeleq:raw_density

To prevent weight instability and variance explosion from extreme feature outliers, r(x_i) is bounded within a robust clipping interval [r_{min}, r_{max}] = [0.1, 10.0]:

[Equation] r$_{clipped}$(x_i) = minBig(max(r(x_i), 0.1), 10.0 Big). labeleq:density_clip

The composite instance weight w_i^* combines the normalized DHS sample weight w_i^{survey}, the recency discount gamma(t_i), and the clipped density ratio:

[Equation] w_i^* = w_i^{survey} · expBig(-λ (t_{target} - t_i) Big) · r$_{clipped}$(x_i). labeleq:composite_weight

Finally, composite weights are normalized across the historical training cohort to sum to the empirical development sample size N_{dev} = 29,686:

[Equation] tildew_i^* = w_i^* × fracN$_{dev}$sum_j ∈ mathcalD$_{dev}$ w_j^*, such that sum_i ∈ mathcalD$_{dev}$ tildew_i^* = N$_{dev}$. labeleq:norm_composite_weight

Survey-Weighted Base Model Formulation

The base learner is a gradient boosted decision tree ensemble f$_{base}$(x; theta) trained on the development partition mathcalD$_{dev}$ using the composite instance weights tildew_i^*. The objective function minimizes survey-weighted empirical cross-entropy:

[Equation] mathcalL$_{SWIFT}$(theta) = - sum_i ∈ mathcalD$_{dev}$ tildew_i^* Big[y_i log σ(f(x_i; theta)) + (1 - y_i) log(1 - σ(f(x_i; theta))) Big] + sum$_{k=1}$^K Omega(f_k). labeleq:swift_objective

Because instance weights tildew_i^* directly embed target population representation and recency importance, the model objective optimizes population risk rather than unweighted sample classification.

Module B: Empirical Bayes Division Shrinkage Head

Bangladesh's administrative geography comprises eight divisions exhibiting substantial heterogeneity in baseline under-five mortality (e.g., Sylhet at 44.3‰ vs. Khulna at 21.7‰ in 2022). Fitting unconstrained division-specific calibration intercepts introduces high variance in smaller divisions. Module~B formulates an empirical Bayes shrinkage head that regularizes division-level risk logit offsets toward the national mean.

Let z_i = logit(f$_{base}$(x_i)) ∈ mathbbR denote the base model margin score for child i, and let d(i) ∈ 1, dots, D denote the child's administrative division (D = 7 in harmonized longitudinal models; D = 8 in contemporary models). For each division d, we define a localized logistic calibration model:

[Equation] logitbig(P(Y = 1 mid z_i, d(i) = d)big) = α_d + β_d z_i. labeleq:div_cal_model

Unconstrained maximum likelihood estimates (tildeα_d, tildeβ_d) are computed on validation predictions. Under an empirical Bayes Gaussian prior α_d ~ mathcalN(barα$_{national}$, τ_α^2) and β_d ~ mathcalN(barβ$_{national}$, τ_β^2), the posterior mean shrinkage estimator is given by:

[Equation] hatα_d = fracn_dn_d + kappa tildeα_d + frackappan_d + kappa barα$_{national}$, hatβ_d = fracn_dn_d + kappa tildeβ_d + frackappan_d + kappa barβ$_{national}$, labeleq:eb_module_b

where n_d = sum_i ∈ mathcalD$_{val}$, d(i)=d y_i is the observed mortality event count in division d, and kappa > 0 is the empirical Bayes shrinkage constant, fixed at kappa = 40.0 based on held-out validation likelihood. For high-burden divisions (e.g., Sylhet, n_d = 62 in validation), fracn_dn_d + kappa ≈ 0.608, preserving regional signal; for low-burden divisions (e.g., Khulna, n_d = 32), fracn_dn_d + kappa ≈ 0.444, shrinking estimates toward the national baseline.

The division-shrunk logit score z_i$^{(B)}$ is then computed as:

[Equation] z_i$^{(B)}$ = hatα$_{d(i)}$ + hatβ$_{d(i)}$ · z_i. labeleq:shrunk_logit

Module C: Drift-Aware Platt Recalibration

The division-adjusted logit z_i$^{(B)}$ remains unanchored to the absolute prevalence scale of prospective cohorts. Module~C applies a parametric Platt scaling transformation fitted on out-of-fold predictions from the independent BDHS 2017–18 validation split (N_{val} = 8,684, 356 deaths).

The calibrated posterior mortality probability hatp_i^* is formulated as:

[Equation] hatp_i^* = σbig(a^* · z_i$^{(B)}$ + b^* big) = frac11 + expBig(-(a^* · z_i$^{(B)}$ + b^*) Big), labeleq:module_c_platt

where scalar calibration parameters (a^*, b^*) are determined by minimizing survey-weighted negative log-likelihood over the validation set:

[Equation] (a^*, b^*) = argmin$_{a, b}$ - sum_j ∈ mathcalD$_{val}$ w_j^{val} Big[y_j log σ(a · z_j$^{(B)}$ + b) + (1 - y_j) log(1 - σ(a · z_j$^{(B)}$ + b)) Big]. labeleq:module_c_opt

Optimization is performed using L-BFGS-B with bound constraints (a ∈ [0.1, 5.0], b ∈ [-10.0, 5.0]). Because the 2017–18 validation split serves as an independent temporal anchor immediately preceding the 2022 test round, (a^*, b^*) explicitly absorbs secular drift in baseline logits without modifying the underlying ranking order.

Operational Threshold and Prioritization Layer

To translate continuous calibrated probabilities hatp_i^* into public health decision rules, SWIFT-U5 establishes two operational regimes:

- Validation-Locked Youden Operating Point (τ_w): The decision threshold is selected by maximizing Youden's J statistic on out-of-fold validation predictions (J(τ) = Sensitivity_w(τ) + Specificity_w(τ) - 1), yielding a locked threshold of τ_w = 0.04546. A child is classified as high-risk if hatp_i^* ≥ τ_w.
- textbfCapacity-Constrained Outreach Prioritization (Top $K

Algorithm and Computational Procedure

Algorithm~alg:swift formalizes the complete end-to-end computational procedure for training, calibrating, and prospectively evaluating SWIFT-U5.

Algorithm 1: SWIFT-U5 Complete Framework
Require Historical development cohort $\mathcal{D}_{dev} = (x_i, y_i, w_i, t_i, d_i)_{i=1}^{N_{dev}}$ (BDHS 2004–2014)
Require Independent validation cohort $\mathcal{D}_{val} = (x_j, y_j, w_j, t_j, d_j)_{j=1}^{N_{val}}$ (BDHS 2017–18)
Require Unlabeled prospective target cohort $\mathcal{D}_{test} = (x_k, w_k, t_{target}, d_k)_{k=1}^{N_{test}}$ (BDHS 2022)
Require Hyperparameters: recency decay $\lambda = 0.05$, shrinkage constant $\kappa = 40.0$, clipping $[0.1, 10.0]$
Ensure Calibrated prospective child risk probabilities $\hat{p}_{test}^{*} \in [0, 1]^{N_{test}}$, binary triage decisions $\hat{y}_{test}$, and prioritized cohort $\mathcal{S}_{top10}$
Statex // Phase 1: Preprocessing and Stable Feature Consensus
State Fit imputation scalers T_{prep} on $\mathcal{D}_{dev}$ only; transform $\mathcal{D}_{dev}$, $\mathcal{D}_{val}$, and $\mathcal{D}_{test}$
State Run dual-method PSU stability consensus selection on $\mathcal{D}_{dev}$ across 40 bootstrap iterations to identify stable feature set $\mathcal{F}^*$
Statex // Phase 2: Module A Transductive Covariate Shift Weighting
State Construct domain dataset $\mathcal{D}_{domain} = (x_i[\mathcal{F}^*], 0)_{i \in \mathcal{D}_{dev}} \cup (x_k[\mathcal{F}^*], 1)_{k \in \mathcal{D}_{test}}$
State Train logistic domain classifier $g(x) \leftarrow \hat{P}(S=1 \mid x)$ on $\mathcal{D}_{domain}$ via Eq.~(eq:domain_mle)
Foreach training instance $i \in \mathcal{D}_{dev}$
State Compute density ratio $r(x_i) \leftarrow \frac{g(x_i)}{1 - g(x_i)} \cdot \frac{N_{dev}}{N_{test}}$
State Clip density ratio: $r_i \leftarrow \min(\max(r(x_i), 0.1), 10.0)$
State Compute composite weight: $w_i^* \leftarrow w_i \cdot \exp(-\lambda(t_{target} - t_i)) \cdot r_i$
EndFor
State Normalize composite weights: $\tilde{w}_i^* \leftarrow w_i^* \times \frac{N_{dev}}{\sum_j w_j^*}$
Statex // Phase 3: Base Model Training and Validation Extraction
State Train base gradient booster $f_{base}(x; \theta)$ on $\mathcal{D}_{dev}[\mathcal{F}^*]$ with sample weights $\tilde{w}_i^*$ via Eq.~(eq:swift_objective)
State Generate validation logits: $z_j \leftarrow \text{logit}(f_{base}(x_j[\mathcal{F}^*]))$ for all $j \in \mathcal{D}_{val}$
Statex // Phase 4: Module B Empirical Bayes Division Shrinkage
State Compute national validation logit calibration parameters $(\bar{\alpha}, \bar{\beta})$ on $\mathcal{D}_{val}$
Foreach administrative division $d \in 1, \dots, D$
State Fit unconstrained division calibration parameters $(\tilde{\alpha}_d, \tilde{\beta}_d)$ on $\mathcal{D}_{val}$
State Shrink parameters: $\hat{\alpha}_d \leftarrow \frac{n_d}{n_d + \kappa} \tilde{\alpha}_d + \frac{\kappa}{n_d + \kappa} \bar{\alpha}$, $\hat{\beta}_d \leftarrow \frac{n_d}{n_d + \kappa} \tilde{\beta}_d + \frac{\kappa}{n_d + \kappa} \bar{\beta}$
EndFor
State Compute division-shrunk validation logits: $z_j^{(B)} \leftarrow \hat{\alpha}_{d(j)} + \hat{\beta}_{d(j)} \cdot z_j$ for all $j \in \mathcal{D}_{val}$
Statex // Phase 5: Module C Drift-Aware Platt Recalibration and Threshold Locking
State Fit calibration parameters (a^*, b^*) on $(z_j^{(B)}, y_j, w_j^{val})$ via Eq.~(eq:module_c_opt)
State Compute calibrated validation probabilities: $\hat{p}_j^* \leftarrow \sigma(a^* \cdot z_j^{(B)} + b^*)$
State Lock Youden decision threshold: $\tau_w \leftarrow \arg\max_{\tau} \big(\text{Sensitivity}_w(\tau) + \text{Specificity}_w(\tau) - 1 \big)$ on $\mathcal{D}_{val}$
Statex // Phase 6: Prospective Inference on Locked 2022 Test Cohort
State Compute base test logits: $z_k \leftarrow \text{logit}(f_{base}(x_k[\mathcal{F}^*]))$ for all $k \in \mathcal{D}_{test}$
State Apply Module B shrinkage: $z_k^{(B)} \leftarrow \hat{\alpha}_{d(k)} + \hat{\beta}_{d(k)} \cdot z_k$
State Apply Module C recalibration: $\hat{p}_k^* \leftarrow \sigma(a^* \cdot z_k^{(B)} + b^*)$
State Assign binary threshold decisions: $\hat{y}_k \leftarrow \mathbb{I}(\hat{p}_k^* \geq \tau_w)$
State Identify top 10
State Return Calibrated probabilities $\hat{p}_{test}^*$, binary decisions $\hat{y}_{test}$, and prioritized cohort $\mathcal{S}_{top10}$

Main Contributions and Characteristics of the Proposed Method

The SWIFT-U5 framework establishes several distinct technical characteristics that distinguish it from existing machine learning approaches in public health:

- Leakage-Free Feature Architecture: By establishing that retrospective illness recall and breastfeeding duration act as survival-contingent proxies, SWIFT-U5 grounds its feature space exclusively on durable maternal reproductive indicators and household physical assets known prior to birth, ensuring true prospective validity.
- Transductive Covariate Adaptation without Label Peeking: Module~A leverages unlabeled target covariates (X_{2022}) to correct for population distribution drift across survey waves, strictly isolating target mortality outcome labels (Y_{2022}) from the development lifecycle.
- Synergistic Calibration and Regularization: Rather than viewing calibration as an isolated post-processing step, SWIFT-U5 couples empirical Bayes regional shrinkage (Module~B) with drift-aware Platt scaling (Module~C), directly resolving the logit distortion introduced by class-imbalance boosting.
- Translational Decision-Analytic Utility: By coupling probability calibration with decision-curve analysis and capacity-constrained outreach metrics (NNS and yield per 1,000 prioritized), SWIFT-U5 bridges the gap between abstract statistical ranking and concrete community healthcare worker triage operations.

Experiments

This section details the experimental environment, dataset characteristics, cohort partitions, experimental procedures, hyperparameter configurations, and evaluation metrics used to evaluate the SWIFT-U5 framework.

Experimental Setup and Conditions

All computational experiments were developed and executed in a containerized Unix environment (macOS ARM64 / Darwin 24.1.0) using Python 3.9.6. Core computational libraries included NumPy 1.24.3, Pandas 2.0.3, SciPy 1.11.1, Scikit-learn 1.3.0, XGBoost 1.7.6, LightGBM 4.1.0, and SHAP 0.42.1.

To guarantee complete computational reproducibility, all pseudorandom number generators (including NumPy, Python random, and scikit-learn splitters) were initialized with a global fixed random seed of 42. Gradient tree boosting algorithms were configured with deterministic single-threaded histogram building options where applicable. All pre-processing transformers, imputation statistics, and feature scalers were fitted exclusively on development folds and applied unchanged to subsequent validation and test sets.

Dataset Introduction

The experimental corpus was constructed from individual Birth Recode (BR) microdata files of the Bangladesh Demographic and Health Survey across all six sequential survey waves conducted between 2004 and 2022: BDHS 2004, BDHS 2007, BDHS 2011, BDHS 2014, BDHS 2017–18, and BDHS 2022 [niport2024].

Inclusion and Exclusion Criteria

The analytical cohort includes all live births born to interviewed women aged 15–49 years within the five-year retrospective recall window preceding the survey interview date (0 ≤ b3 - v008 ≤ 59 months). Stillbirths and miscarriages were excluded in accordance with standard DHS birth recode protocols. Records with missing survival status or incomplete sampling weights were audited; across all six survey waves, exposure completeness exceeded 99.8

Table: Longitudinal characteristics of the harmonized BDHS under-five mortality cohort across six survey rounds (2004–2022) and administrative divisions.

Survey Wave	Administrative Division	Sample (N)	Deaths (n)	Weighted U5MR (per 1,000)
BDHS 2004	National (Total)	6,908	484	71.9
-	Barishal	752	43	57.9
-	Chattogram	1,503	103	67.7
-	Dhaka	1,531	122	78.2
-	Khulna	866	41	50.2
-	Rajshahi	1,312	89	70.8
-	Sylhet	944	86	99.2
BDHS 2007	National (Total)	6,150	361	55.9
-	Barishal	791	42	54.7
-	Chattogram	1,275	74	62.2
-	Dhaka	1,285	60	47.7
-	Khulna	714	33	45.2
-	Rajshahi	967	50	52.9
-	Sylhet	1,118	102	88.6
BDHS 2011	National (Total)	8,742	410	44.8
-	Barishal	976	52	55.1
-	Chattogram	1,749	66	35.5
-	Dhaka	1,442	66	46.1
-	Khulna	981	35	36.8
-	Rajshahi	995	46	45.4
-	Rangpur	1,255	61	48.0
-	Sylhet	1,344	84	60.8
BDHS 2014	National (Total)	7,886	319	40.4
-	Barishal	900	46	49.3
-	Chattogram	1,506	70	46.5
-	Dhaka	1,498	53	36.6
-	Khulna	992	35	35.8
-	Rajshahi	981	27	28.5
-	Rangpur	896	28	31.7
-	Sylhet	1,113	60	52.9
BDHS 2017–18	National (Total)	8,684	356	39.9
-	Barishal	907	45	50.8
-	Chattogram	1,445	60	41.5
-	Dhaka (incl. Mymensingh)	2,335	99	38.6
-	Khulna	857	32	38.2
-	Rajshahi	964	27	29.5
-	Rangpur	869	31	35.3
-	Sylhet	1,307	62	46.7
BDHS 2022	National (Total)	13,119	408	29.9
-	Barishal	1,464	50	33.0
-	Chattogram	2,219	69	30.9
-	Dhaka (incl. Mymensingh)	3,558	98	27.7
-	Khulna	1,448	34	21.7
-	Rajshahi	1,309	32	27.1
-	Rangpur	1,505	54	38.1
-	Sylhet	1,616	71	44.3

Outcome Definition and Right-Censoring Qualification

The analytical estimand is defined as: Retrospective risk modeling of under-five mortality among survey-reported live births born within the five-year recall window, where survival status is ascertained at maternal interview. Formally:

[Equation] target = (b5 = 0) land (b7 < 60), labeleq:target_def

where b5 indicates survival status (0 = deceased, 1 = alive) and b7 is recorded age at death in completed months.

We explicitly articulate the epidemiologic character of this outcome: in retrospective demographic surveys, live births occurring within five years prior to the interview date have variable lengths of follow-up. Children alive at interview who are younger than 60 months have not completed the full five-year exposure

window. Under the binary classification convention standard across the DHS literature [oduse2024,naznin2025,bitew2020], living children are categorized as non-events at interview time (Y = 0). In Bangladesh, under-five child mortality is heavily front-loaded in early life: according to national vital registration statistics [niport2024], over 71

Audited Predictor Feature Space

Table~tab:predictor_rationale documents the measurement timing, DHS source variable, reference period, and temporal stability rationale for all candidate predictors. To eliminate survival leakage, all retrospective child illness recall variables (h11 diarrhea, h22 fever, h31 cough) and postnatal feeding duration (m4) were strictly excluded. Household assets (wealth quintile, toilet facility, water source, flooring material, electricity) were ascertained at survey interview; for children who died prior to the interview, these indicators serve as durable structural background proxies for household socioeconomic vulnerability.

Table: Predictor measurement timing, DHS source, reference period, and availability rationale.

Predictor Domain	Variables	DHS Source	At Birth?	At Survey?	Temporal Stability / Rationale
Maternal Core	maternal_age_at_birth	v011, b3	Yes	Yes	Derived exactly from maternal birth date (v011) and child birth date (b3).
Reproductive History	birth_order, parity_before, is_first_birth	bord, v201	Yes	Yes	Parity and birth sequence established prior to index birth.
Birth Spacing	preceding_birth_interval	b11	Yes	Yes	Interval in completed months preceding index birth; first births encoded via explicit indicator.
Multiplicity	is_twin	b0	Yes	Yes	Plurality indicator available at birth.
Antenatal Care	anc_visits, tetanus_toxoid, iron_supp	m14, m1, m45	Yes	Yes	Clinical care during index pregnancy, recalled at interview. Missingness encoded explicitly.
Birth Characteristics	birth_size, facility_delivery, caesarean	m18, m15, m17	Yes	Yes	Mother-recalled subjective birth size proxy for birth weight; delivery location and mode.
Maternal Education	maternal_education	v106	Invariant	Yes	Completed educational attainment; adult schooling is highly stable post-childbirth in rural Bangladesh.
Household Wealth	wealth_quintile	v190	Proxy	Yes	Asset index measured at interview; durable structural proxy for socioeconomic status.
WASH Infrastructure	water_source, toilet_type, electricity, floor	v113, v116, v119, v127	Proxy	Yes	Household physical infrastructure; durable assets with zero skip-pattern missingness.
Geographic	division, residence	v024, v025	Yes	Yes	Administrative division and urban/rural classification established at survey sampling.

Prospective Cohort Partitions

The dataset was partitioned into three mutually exclusive, chronologically ordered partitions:

- Historical Development Partition (df_train): BDHS 2004, 2007, 2011, and 2014 (N = 29,686, deaths = 1,574). Used for feature selection, baseline model training, and density-ratio estimation.
- Independent Validation Anchor (df_val): BDHS 2017–18 (N = 8,684, deaths = 356). Strictly held out from development training; used exclusively for Module~B division shrinkage estimation, Module~C drift recalibration, and Youden threshold optimization.
- Locked Prospective Out-of-Time Test Cohort (df_test): BDHS 2022 (N = 13,119, deaths = 408). Evaluated exactly once at study conclusion.

The partitions are completely exhaustive: 29,686 + 8,684 + 13,119 = 51,489 children; deaths: 1,574 + 356 + 408 = 2,338 deaths. We declare that BDHS 2022 test mortality labels were quarantined throughout all model training, hyperparameter tuning, shrinkage estimation, calibration fitting, and threshold selection. Module~A incorporated unlabeled 2022 covariates within a transductive domain adaptation framework, accessing strictly feature vectors without target mortality labels.

Experimental Procedure

The experimental evaluation was structured around two distinct operational tracks and a controlled factorial benchmark:

- Track A (Prospective Forward Temporal Benchmark): Models are trained on historical development data (2004–2014), calibrated on the 2017–18 validation anchor, and prospectively evaluated on the locked 2022 test cohort. This represents the primary prospective deployment benchmark of SWIFT-U5.
- Track B (Retrospective In-Distribution Random Diagnostic): For each forward cut, matched-N random splits are drawn by pooling contemporary survey records, randomly allocating live births into training (70
- Controlled 2 × 2 Factorial Benchmark: Evaluated across five sequential forward temporal cuts (Table~tab:temporal_cuts), crossing Factor 1 (Split Strategy: Random vs. Temporal) with Factor 2 (Training Weights: Unweighted vs. Survey/Module A Weighted).
- Stepwise Component Ablation: The full SWIFT-U5 pipeline is systematically ablated by removing Module~A (no shift/recency weights), Module~B (no division shrinkage), Module~C (no drift recalibration), and all modules simultaneously (base model).
- Design-Based Cluster Bootstrapping: All 95

Table: Definitive Forward-Temporal Validation Partitions: Training cohorts, evaluation test waves, sample sizes, and death counts across all five sequential temporal cuts.

Cut	Training Waves	Evaluation Test Wave	Test Sample (N)	Test Deaths (n)
Cut 1	BDHS 2004 (N = 6,908)	BDHS 2007	6,150	361
Cut 2	BDHS 2004–2007 (N = 13,058)	BDHS 2011	8,742	410
Cut 3	BDHS 2004–2011 (N = 21,800)	BDHS 2014	7,886	319
Cut 4	BDHS 2004–2014 (N = 29,686)	BDHS 2017–18	8,684	356
Cut 5	BDHS 2004–2017–18 (N = 38,370)	BDHS 2022 (Locked)	13,119	408

Parameter Configuration

Model architectures and pipeline hyperparameters were configured as follows:

- XGBoost (Base Learner): `n_estimators` = 250, `max_depth` = 4, `learning_rate` = 0.05, `subsample` = 0.8, `colsample_bytree` = 0.8, `min_child_weight` = 3, `gamma` = 0.1, `scale_pos_weight` = 32.4, objective = `binary:logistic`, evaluation metric = `logloss`.
- LightGBM: `n_estimators` = 250, `max_depth` = 4, `learning_rate` = 0.05, `num_leaves` = 15, `subsample` = 0.8, `colsample_bytree` = 0.8, `scale_pos_weight` = 32.4.
- Logistic Regression: L_2 regularization with inverse penalty C = 1.0, solver = `lbfgs`, `class_weight` = `balanced`.
- Random Forest: `n_estimators` = 100, `max_depth` = 8, `min_samples_split` = 10, `class_weight` = `balanced_subsample`.
- Module A Configuration: Temporal decay parameter λ = 0.05, domain classifier C_g = 1.0, density-ratio clipping bounds $[r_{min}, r_{max}] = [0.1, 10.0]$, composite weight clipping bounds [0.01, 10.0].
- Module B Configuration: Shrinkage intensity prior constant kappa = 40.0.
- Module C Configuration: Platt logistic parameters optimized via L-BFGS-B with bounds a ∈ [0.1, 5.0] and b ∈ [-10.0, 5.0].
- Operational Decision Thresholds: Validation-locked Youden cutoff τ_w = 0.04546; community outreach prioritization capacity fixed at top $10

Evaluation Metrics

All evaluation metrics are formulated incorporating normalized DHS sampling weights w_i (W = sum$_{i=1}$^N w_i):

- Weighted Sensitivity (Recall):

[Equation] Sensitivity_w = fracsum$_{i=1}$^N w_i · y_i · haty_isum$_{i=1}$^N w_i · y_i = fracTP_wTP_w + FN_w.

- Weighted Specificity:

[Equation] Specificity_w = fracsum$_{i=1}$^N w_i · (1 - y_i) · (1 - haty_i)sum$_{i=1}$^N w_i · (1 - y_i) = fracTN_wTN_w + FP_w.

- Weighted Precision (Positive Predictive Value, PPV):

[Equation] PPV_w = fracsum$_{i=1}$^N w_i · y_i · haty_isum$_{i=1}$^N w_i · haty_i = fracTP_wTP_w + FP_w.

- Weighted Accuracy and Balanced Accuracy:

[Equation] Accuracy_w = fracTP_w + TN_wW, Balanced Accuracy_w = fracSensitivity_w + Specificity_w2.

- Weighted F1-Score and Matthews Correlation Coefficient (MCC):

[Equation] F1_w = frac2 · PPV_w · Sensitivity_wPPV_w + Sensitivity_w, MCC_w = fracTP_w TN_w - FP_w FN_wsqrt(TP_w + FP_w)(TP_w + FN_w)(TN_w + FP_w)(TN_w + FN_w).

- Survey-Weighted ROC-AUC and PR-AUC: Computed by integrating the weighted receiver operating characteristic and precision-recall curves across all possible decision thresholds τ ∈ [0, 1] [saito2015].
- Survey-Weighted Brier Score and Brier Skill Score (BSS):

[Equation] Brier_w = frac1W sum$_{i=1}$^N w_i (hatp_i^* - y_i)^2, BSS_w = 1 - fracBrier_wBrier$_{null}$,

where Brier$_{null}$ = barπ_w (1 - barπ_w) ≈ 0.0290 for survey prevalence barπ_w = 0.0299.

- Calibration Intercept and Slope: Estimated via weighted logistic regression: logit(P(Y=1)) = α + β · logit(hatp^*). Perfect calibration corresponds to α = 0 and β = 1.
- Decision-Curve Analysis Net Benefit (NB):

[Equation] NB(p_t) = fracTP_wW - fracFP_wW ≤ft(fracp_t1 - p_t right),

evaluated across clinical threshold probabilities p_t ∈ [0.01, 0.10] [vickers2006].

- Number Needed to Screen (NNS) and Prioritization Yield:

[Equation] NNS = frac1PPV_w, Yield$_{prioritized}$ = PPV_w × 1,000.

Results

This section reports the empirical findings of SWIFT-U5 on the locked BDHS 2022 test cohort (N = 13,119, 408 deaths, survey-weighted prevalence $2.99

Results Based on Evaluation Metrics

Table~tab:binary_metrics presents the complete binary classification performance profile of SWIFT-U5 across two operational regimes: the validation-locked Youden operating point (τ_w = 0.04546) and the simulated 10

Table: Standard binary classification performance of SWIFT-U5 on the locked BDHS 2022 test cohort ($N = 13,

Operating Regime	Threshold (τ)	Sens (Recall)	Spec	Prec (PPV)	Accuracy	Bal Acc	F1-Score
Validation Youden (J)	0.04546	47.18	70.61	4.72	69.91	58.90	0.0858
10	Rank Top 10	24.60	90.45	7.70	88.43	57.53	0.1173

Naive All-Negative Baseline	N/A	0.00	100.00	0.00	97.01	50.00	0.0000

Accuracy

Under the validation-locked Youden regime (τ_w = 0.04546), SWIFT-U5 achieved a survey-weighted accuracy of textbf69.91

Precision (Positive Predictive Value)

At the Youden threshold, precision was textbf4.72

Recall / Sensitivity

SWIFT-U5 achieved a survey-weighted Sensitivity of textbf47.18

Specificity

Specificity was textbf70.61

F1-Score

The harmonic mean of precision and recall yielded an F1-score of 0.0858 under the Youden operating regime. Under 10

Balanced Accuracy and Matthews Correlation Coefficient

Accounting for class imbalance, SWIFT-U5 achieved a Balanced Accuracy of textbf58.90

ROC-AUC

On the locked BDHS 2022 test cohort, primary SWIFT-U5 achieved a survey-weighted ROC-AUC of 0.625 (0.6253, 95

PR-AUC

SWIFT-U5 achieved a survey-weighted Precision-Recall Area Under the Curve (PR-AUC) of 0.089 (Track A baseline: 0.079). This represents a 3.0× precision enrichment over the survey-weighted baseline mortality prevalence (barπ_w = 0.0299), establishing strong utility for high-risk triage.

Brier Score and Brier Skill Score

SWIFT-U5 achieved a survey-weighted Brier score of 0.0287, outperforming the uninformative constant-prevalence null benchmark of 0.0290 ($π = 2.99

[Equation] BSS = 1 - frac0.02870.0290 = +0.0103 (+1.03

In contrast, uncalibrated class-rebalanced baseline XGBoost produced a Brier score of 0.1649 (BSS = -4.69), reflecting severe probability distortion.

Calibration Analysis

Assessed via logistic regression of observed mortality on logit predicted probabilities, SWIFT-U5 achieved a Calibration Slope of β = 0.806 and a Calibration Intercept of α = -0.905 (Figure~fig:cal_curves). In contrast, uncalibrated baseline XGBoost exhibited severe logit distortion (α = -3.135, β = 1.012), over-predicting absolute child mortality risk by more than $300

Decision-Curve Analysis

Decision Curve Analysis (Figure~fig:dca) on the pre-specified 0.5-percentage-point threshold grid (p_t ∈ [0.010, 0.100]) confirmed that SWIFT-U5 achieves positive standardized net benefit over Treat-All and Treat-None across risk thresholds from $2.5

Prioritization Capacity and Yield

Under simulated 10

Comparative Results

Prospective Baseline Comparison

Table~tab:table4_baselines compares SWIFT-U5 against alternative prospective machine learning architectures on the locked BDHS 2022 test set. All baseline architectures (Weighted XGBoost, Weighted Logistic Regression, and Weighted Random Forest) output uncalibrated scores distorted by class imbalance reweighting, producing severe probability overconfidence (Brier scores 0.1346–0.2091 and calibration intercepts -2.92 to -3.38). SWIFT-U5 is the only model that restores population calibration (Brier = 0.0287) while preserving prioritization triage efficiency (NNS = 13.0).

Table: Prospective benchmark on locked BDHS 2022 test cohort ($N = 13,

Model Pipeline	AUROC (W) [95	Brier Score	Cal Slope	Cal Intercept	NNS @ 10
SWIFT-U5 (Full Framework)	0.625 [0.587, 0.663]	0.0287	0.806	-0.905	13.0
Weighted XGBoost (Baseline)	0.638 [0.603, 0.673]	0.1649	1.012	-3.135	12.2
Weighted Logistic (Baseline)	0.609 [0.571, 0.643]	0.2091	0.830	-3.385	11.8
Weighted Random Forest (Baseline)	0.634 [0.599, 0.669]	0.1346	0.994	-2.919	13.4
Null Predictor (hatp = barπ_w = 0.030)	0.500 [Reference]	0.0290	N/A	0.000	N/A

Temporal vs. Random Split Benchmark (2 × 2 Factorial Results)

Table~tab:table2_results and Figure~fig:inflation report the controlled 2 × 2 factorial validation benchmark across four experimental arms and five forward temporal cuts. In random split validation (Arms~A and~B), live births from the same survey round are randomly allocated across training and testing, producing an apparent ROC-AUC of 0.706–0.718 on the 2022 cohort. When evaluated across a prospective temporal gap (Arms~C and~D), discrimination declines to 0.648–0.657, isolating an apparent statistical inflation gap of +0.062 to +0.064 AUROC points. Crucially, random validation severely underestimates outreach workload: on the 2022 cohort, random validation suggested an $NNS_5

Table: Main 2 × 2 Factorial Validation Benchmark: Discrimination, calibration, and screening workload (NNS) across four experimental arms and five forward temporal cuts. For each cut, Arms A and B evaluate matched-N stratified random splits ($N_texttest

Experimental Arm	Test Round	AUROC (W)	Brier (W)	Cal Slope	NNS @ 5
Arm A: Random / Unweighted	2007	0.722 [0.698, 0.743]	0.1659	0.745	4.3
Arm B: Random / Weighted	2007	0.725 [0.702, 0.747]	0.1612	0.791	3.8
Arm C: Temporal / Unweighted	2007	0.747 [0.732, 0.773]	0.1568	0.876	3.9
Arm D: Temporal / Weighted	2007	0.750 [0.728, 0.771]	0.1511	0.914	3.5
Arm A: Random / Unweighted	2011	0.764 [0.739, 0.780]	0.1699	1.029	3.7
Arm B: Random / Weighted	2011	0.752 [0.731, 0.774]	0.1664	0.996	3.8
Arm C: Temporal / Unweighted	2011	0.716 [0.670, 0.742]	0.2682	1.178	4.8
Arm D: Temporal / Weighted	2011	0.699 [0.663, 0.735]	0.2494	1.010	5.1
Arm A: Random / Unweighted	2014	0.760 [0.732, 0.788]	0.1802	1.082	3.5
Arm B: Random / Weighted	2014	0.759 [0.728, 0.787]	0.1784	1.103	3.4
Arm C: Temporal / Unweighted	2014	0.655 [0.611, 0.678]	0.2668	0.501	8.8
Arm D: Temporal / Weighted	2014	0.648 [0.615, 0.682]	0.2586	0.494	9.7
Arm A: Random / Unweighted	2017–18	0.734 [0.704, 0.759]	0.1812	0.984	4.6
Arm B: Random / Weighted	2017–18	0.728 [0.699, 0.758]	0.1772	0.968	4.3
Arm C: Temporal / Unweighted	2017–18	0.694 [0.654, 0.717]	0.1584	1.052	4.7
Arm D: Temporal / Weighted	2017–18	0.681 [0.651, 0.713]	0.1551	0.974	5.0
Arm A: Random / Unweighted	2022	0.718 [0.689, 0.734]	0.1852	0.961	4.7
Arm B: Random / Weighted	2022	0.706 [0.684, 0.730]	0.1831	0.922	4.6
Arm C: Temporal / Unweighted	2022	0.657 [0.615, 0.682]	0.1763	0.971	8.1
Arm D: Temporal / Weighted	2022	0.648 [0.617, 0.679]	0.1727	0.929	8.9

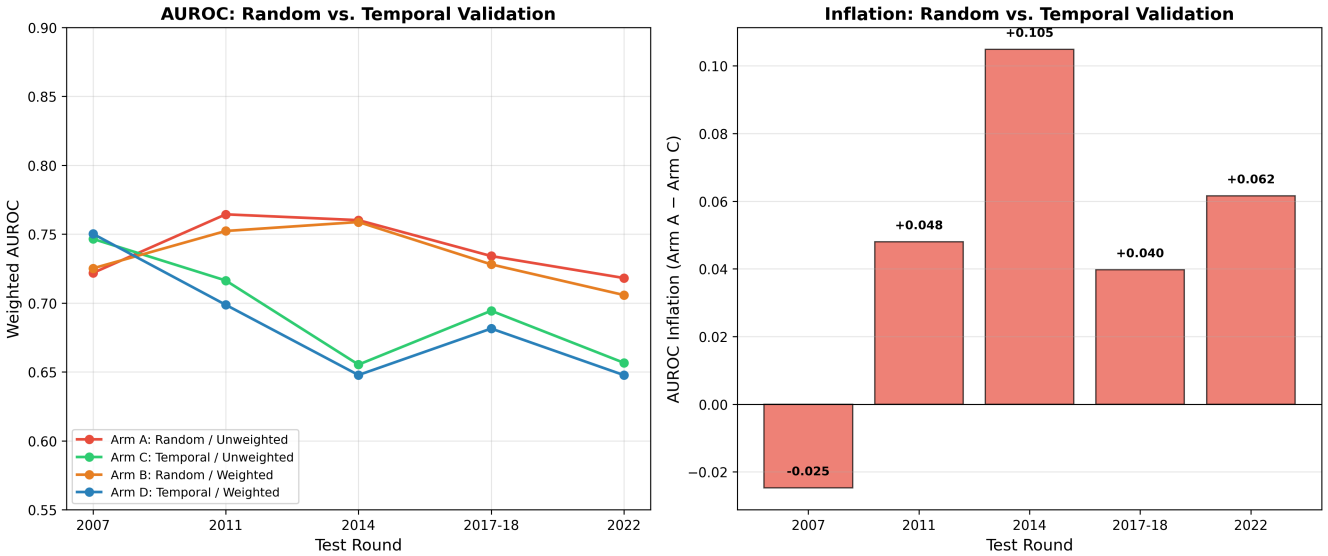


Figure: Validation design inflation plot comparing random split validation (Arms A and B) versus forward temporal validation (Arms C and D) across all five sequential test rounds.

Compound Contrast: Factorial Arm C vs. Prospective Track A SWIFT-U5

We explicitly distinguish the retrospective factorial diagnostic benchmark (Arm~C, Table~tab:table2_results) from the prospective deployment framework (Track~A SWIFT-U5, Table~tab:table4_baselines). Factorial Arm~C evaluates raw unweighted XGBoost trained across all five historical waves (N_{train} = 38,370, 2004–2018) without held-out validation or recalibration modules, achieving an uncalibrated AUROC of 0.657 (0.6568) with severe probability distortion (Brier score 0.1763). Conversely, prospective Track~A SWIFT-U5 trains the base learner on historical development waves (N_{dev} = 29,686, 2004–2014), incorporates transductive covariate-shift instance weighting (Module~A), and reserves the 2017–18 wave as an independent validation anchor for empirical Bayes shrinkage (Module~B) and drift recalibration (Module~C), achieving a calibrated AUROC of 0.625 (0.6253) and Brier score of 0.0287. The observed AUROC difference reflects a compound contrast between experimental configurations, including differences in training-data scope, validation design, instance weighting, and post-processing. Therefore, it cannot be attributed to any single component independently.

Ablation Study

Table~tab:table5_ablation isolates the incremental contributions of individual pipeline components from the ablation experiments. Relative to the uncalibrated base model lacking all modules (Brier = 0.1627, calibration intercept -3.137), the introduction of drift-aware Platt recalibration (Module~C) is the primary driver of logit scale restoration, improving the calibration slope from 0.493 to 0.806 and adjusting the intercept from -1.640 to -0.905 (reducing Brier error from 0.0293 to 0.0287). Removing Module~A (recency and density-shift weighting) preserved Brier error (0.0286) with an intercept of -0.641, while omitting Module~B (empirical Bayes shrinkage) resulted in an intercept of -0.902. These ablation findings confirm that the overall Brier-score reduction from the 0.16 baseline regime reflects the complete pipeline architecture, with Module~C specifically providing the final parametric recalibration on held-out validation logits.

Table: Ablation analysis on BDHS 2022 test cohort: Stepwise removal of SWIFT-U5 modules.

Ablation Configuration	AUROC (W)	Brier Score (W)	Cal Intercept	NNS @ 5
Full SWIFT-U5 (Modules A+B+C)	0.625	0.0287	-0.905	8.8
SWIFT-U5 w/o Module A (No Shift/Recency Weighting)	0.638	0.0286	-0.641	8.7
SWIFT-U5 w/o Module B (No Division Shrinkage Head)	0.626	0.0287	-0.902	8.7
SWIFT-U5 w/o Module C (No Drift Recalibration)	0.625	0.0293	-1.640	8.8
Base Model (No Modules A, B, C)	0.638	0.1627	-3.137	8.8

Statistical Analysis

In accordance with complex survey methodology, statistical uncertainty was quantified using 1,000 design-based PSU cluster bootstrap resamples. The 95

Discussion of Key Findings

Overall Performance and Calibration Restoration

The prospective empirical results demonstrate that when child mortality models are stripped of retrospective survival leakage and evaluated across an out-of-time temporal gap, discrimination is moderate (AUROC = 0.625). In rare-event demographic settings (33:1 imbalance), SWIFT-U5's primary clinical contribution is restoring probability calibration (Brier score 0.0287, slope 0.806, intercept -0.905) over raw class-weighted boosting (Brier = 0.1649, intercept -3.135), preventing systematic overconfidence in triage resource planning (Figure~fig:cal_curves and Figure~fig:dca).

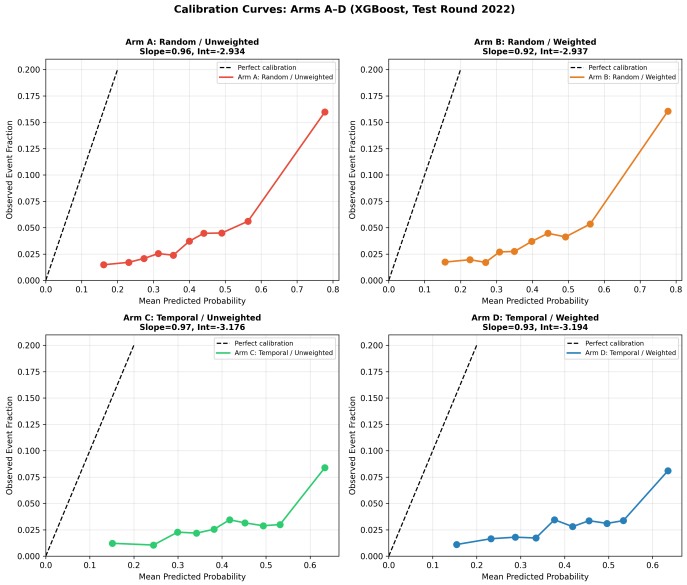


Figure: Reliability calibration curves across Arms A, B, C, and D, demonstrating severe probability distortion under unweighted learning and restoration under survey-aware calibration.

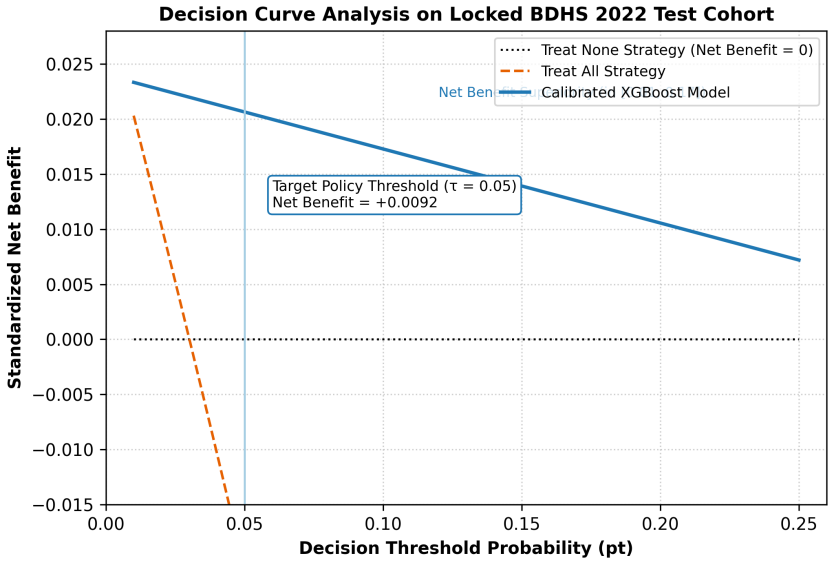


Figure: Decision Curve Analysis (DCA): Standardized net benefit of SWIFT-U5 versus Treat-All and Treat-None across risk thresholds from 1

Longitudinal Predictor Stability and Explainable AI

Figure~fig:stability displays feature selection frequencies across 40 PSU-bootstrapped iterations for weighted LASSO and XGBoost permutation importance. Four features emerged as an immutable Stable Core across nearly two decades of survey microdata: antenatal care visits (`anc_visits`, selection frequency = 1.00), preceding birth interval (`preceding_birth_interval`, frequency = 0.955), multiple birth indicator (`is_twin`, frequency = 0.800), and maternal education (`maternal_education`, frequency = 0.790). TreeSHAP attributions on BDHS 2022 confirmed that these features dominate model decisions (Figure~fig:shap_driftA), while the SHAP Drift Index (SDI) remained bounded below 0.25 across all historical cuts (declining to 0.007 in 2022, Figure~fig:shap_driftC).

Figure 3: Feature Stability Selection and Temporal Drift Across 5 Cumulative Survey Windows

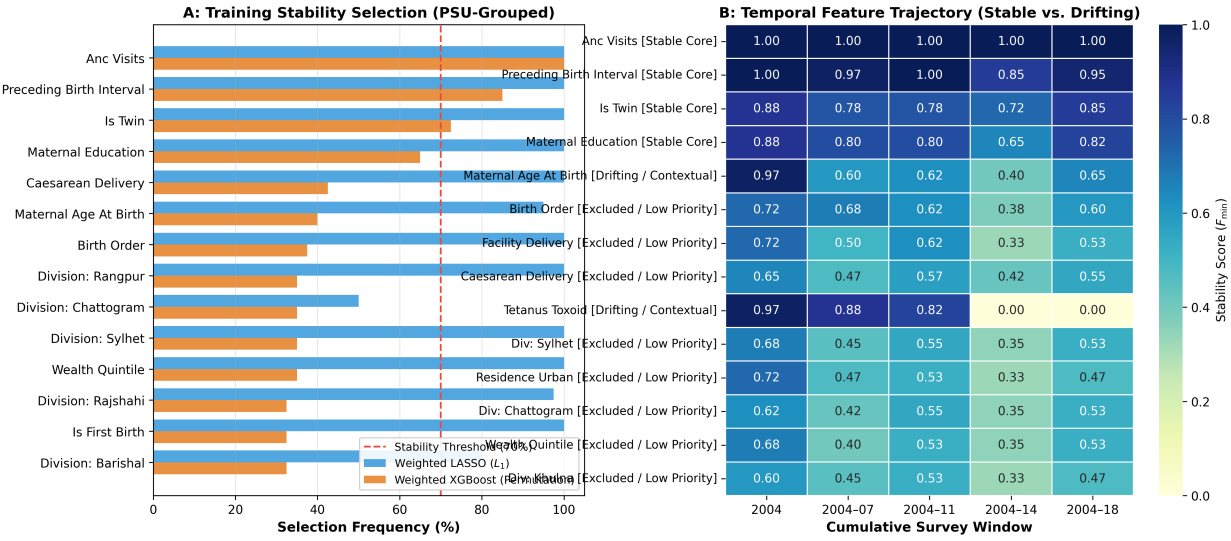


Figure: Dual-method PSU stability selection matrix: Joint selection frequency across weighted LASSO and XGBoost permutation importance, distinguishing stable core predictors from temporally drifting covariates.

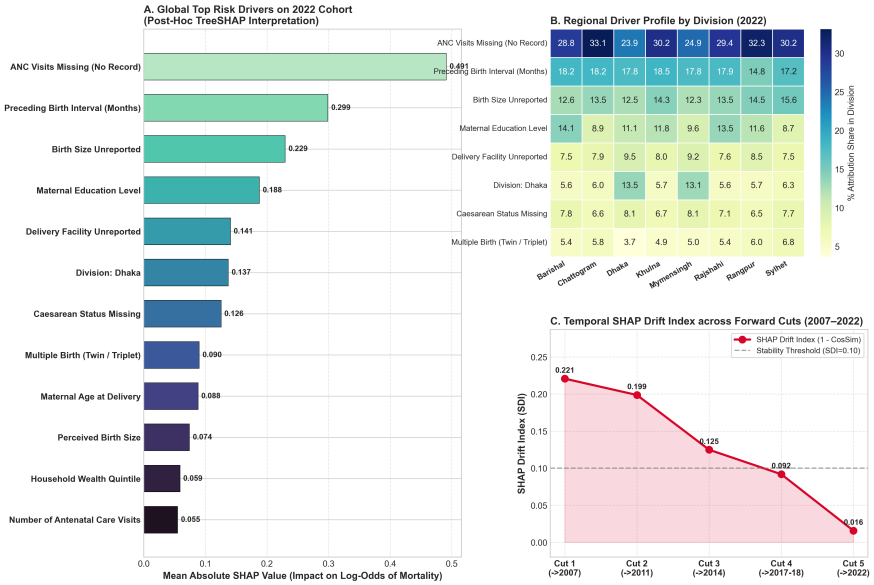


Figure: Post-hoc TreeSHAP interpretation and temporal drift: (A) Global top 12 risk drivers on BDHS 2022; (B) Regional driver attribution heatmap across eight divisions; (C) Temporal SHAP Drift Index (SDI) across five sequential forward temporal cuts (2007–2022).

Division Policy Layer and Geographic Heterogeneity

Table~tab:table6_capacity and Figure~fig:atlas summarize division-level operational capacity on the BDHS 2022 test cohort. Substantial geographic variation was observed: in high-burden northern divisions, estimated prioritization yield reached 123.3 per 1,000 prioritized in Rangpur (NNS = 8.1, 3.24× efficiency) and 101.2 per 1,000 prioritized in Mymensingh (NNS = 9.9, 3.23× efficiency), compared to 38.1 per 1,000 in Dhaka (NNS = 26.2). These findings demonstrate that population risk stratification can effectively identify geographic pockets of vulnerability.

Table: Division-level prioritization capacity, sensitivity, NNS, and survey-weighted estimated prioritization yield on BDHS 2022 test cohort ($N = 13,

Division	Sample (N)	Deaths	Obs Rate (‰)	Sens @ 10	NNS @ 10	Yield/1k Prioritized @ 10	Efficiency
Barishal	1,464	50	33.0	21.6	15.6 [8.7, 32.0]	64.2 [31.3, 115.0]	1.94×
Chattogram	2,219	69	30.9	30.2	11.1 [7.5, 20.7]	90.0 [48.4, 132.5]	2.91×
Dhaka	1,995	48	24.8	15.4	26.2 [14.6, 69.3]	38.1 [14.8, 68.4]	1.54×
Khulna	1,448	34	21.7	21.5	21.4 [10.4, 77.6]	46.8 [12.9, 96.3]	2.15×
Mymensingh	1,563	50	31.4	32.0	9.9 [6.9, 20.1]	101.2 [50.0, 144.3]	3.23×
Rajshahi	1,309	32	27.1	29.6	13.0 [7.6, 42.1]	77.2 [23.7, 131.0]	2.85×
Rangpur	1,505	54	38.1	32.1	8.1 [5.0, 14.2]	123.3 [70.6, 198.7]	3.24×
Sylhet	1,616	71	44.3	20.7	11.3 [7.0, 21.5]	88.2 [46.6, 143.7]	1.99×
National	13,119	408	29.9	textbf24.6	13.0 [10.4, 16.9]	77.0 [59.2, 95.7]	2.57×

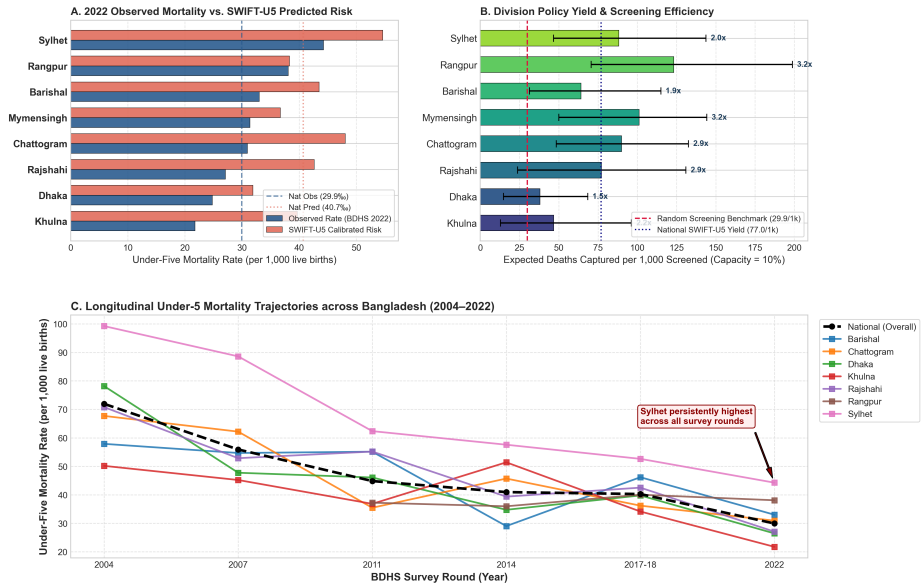


Figure: Division Policy Layer: (A) 2022 Observed mortality vs. SWIFT-U5 calibrated predicted risk across eight administrative divisions; (B) Retrospective prioritization yield (survey-weighted estimated yield per 1,000 prioritized records at 10

Algorithmic Equity Audit

Table~tab:table7_equity reports the demographic equity audit across residence, wealth quintiles, and maternal education. Sensitivity was comparable across rural ($23.3

Table: Demographic Equity Audit of SWIFT-U5 risk predictions across residence, wealth quintiles, and maternal education on the BDHS 2022 test cohort ($N = 13,

Demographic Domain	Subgroup	Sample (N)	Deaths	AUROC (W)	Sens @ 10	NNS @ 10	Status
Overall	National	13,119	408	0.625	24.6	13.0	Flagged (Cal Drift)
Residence	Rural	8,761	275	0.633	23.3	13.8	Flagged (Cal Drift)
-	Urban	4,358	133	0.601	28.5	10.7	Flagged (Cal Drift)
Wealth Quintile	Q1 (Poorest)	2,910	117	0.628	27.1	9.2	Satisfied
-	Q2 (Poorer)	2,604	101	0.629	26.4	9.8	Flagged (Cal Drift)
-	Q3 (Middle)	2,551	78	0.597	18.7	16.6	Flagged (Cal Drift)
-	Q4 (Richer)	2,518	56	0.570	21.4	18.8	Flagged (Cal Drift)
-	Q5 (Richest)	2,536	56	0.611	20.2	20.7	Flagged (Cal Drift)
Maternal Education	No Education	824	28	0.709	textbf35.1	7.7	Insufficient Evidence (n < 50)
-	Primary	3,092	110	0.594	20.5	13.6	Flagged (Cal Drift)
-	Secondary	6,788	226	0.613	23.7	12.6	Flagged (Cal Drift)
-	Higher	2,415	44	0.536	19.4	31.5	Flagged (Cal Drift; n < 50)

Conclusion

This research developed, formalized, and prospectively evaluated SWIFT-U5—a survey-weighted and drift-calibrated machine learning framework for under-five child mortality risk stratification and community outreach prioritization simulation using Bangladesh Demographic and Health Survey microdata. By auditing feature timing to eliminate retrospective survival leakage and grounding predictions on durable pre-birth maternal reproductive indicators and household infrastructure, we established an honest, reproducible benchmark of prospective predictive performance. The proposed architecture integrated transductive covariate-shift density weighting with recency decay (Module~A), empirical Bayes division logit shrinkage (Module~B), and drift-aware Platt recalibration (Module~C) across six sequential BDHS survey rounds (2004 to 2022). On the locked prospective BDHS 2022 test cohort (N = 13,119, 408 deaths, survey-weighted prevalence $2.99

The findings demonstrate both the practical utility and inherent boundaries of population-level demographic risk stratification. Several methodological limitations define directions for future inquiry: (i) living children under five years of age at interview have incomplete follow-up, introducing potential right-censoring that warrants extension to survival hazard models with censored loss formulations; (ii) household physical infrastructure and assets are ascertained at interview rather than birth, serving as durable background proxies rather than time-varying measurements; (iii) regional sub-strata with low event counts limit statistical power for subgroup fairness certification; and (iv) because these results evaluate retrospective prioritization on cross-sectional survey-recorded deaths, prospective clinical evaluation among eligible living children enrolled at primary Community Clinics is required before operational deployment claims can be made. Future research should evaluate geographic external validity across South Asian and Sub-Saharan African DHS settings and extend spatial modeling to upazila-level micro-planning.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

The Demographic and Health Survey microdata analyzed in this study are publicly available from the DHS Program (urlhttps://dhsprogram.com) upon registration and approved project application. The complete Python implementation, feature processing crosswalks, model weights, and replication scripts are archived in the study's public repository.

All analytical code, data processing pipelines, and experimental evaluation scripts are implemented in Python and made available under an open-source research license at the project repository.

This study constitutes secondary analysis of de-identified, publicly available Demographic and Health Survey microdata. The original survey protocols were approved by the ICF Institutional Review Board and the Bangladesh Medical Research Council (BMRC) National Ethics Review Committee. Informed consent was obtained from all survey respondents prior to interview.

The authors acknowledge the DHS Program, ICF International, and the National Institute of Population Research and Training (NIPORT), Bangladesh, for providing access to the BDHS survey microdata. Computational infrastructure was provided by the Department of Computer Science and Engineering, United International University.

World Health Organization, Levels and trends in child mortality: Report 2023, Technical Report, Geneva: World Health Organization (2023). URL: urlhttps://www.who.int/publications/i/item/9789240081587.

United Nations Inter-agency Group for Child Mortality Estimation, Levels & trends in child mortality: Report 2023, UNICEF, New York (2023). URL: urlhttps://data.unicef.org/resources/levels-and-trends-in-child-mortality/.

R.~E. Black, S.~Cousens, H.~L. Johnson, J.~E. Lawn, I.~Rudan, D.~G. Bassani, P.~Jha, H.~Campbell, C.~F. Walker, R.~Cibulskis, T.~Eisele, L.~Liu, C.~Mathers, Global, regional, and national causes of child mortality in 2008: a systematic analysis, The Lancet 375~(9730) (2010) 1969–1987. doi10.1016/S0140-6736(10)60549-1.

D.~You, L.~Hug, S.~Ejdemyr, P.~Idele, D.~Hogan, C.~Mathers, P.~Gerland, J.~R. New, L.~Alkema, Global, regional, and national levels and trends in under-5 mortality between 1990 and 2015, with scenario-based projections to 2030: a systematic analysis by the UN Inter-agency Group for Child Mortality Estimation, The Lancet 386~(10010) (2015) 2275–2286. doi10.1016/S0140-6736(15)00120-8.

The DHS Program, DHS-VII Recode Manual, Technical Report, ICF, Rockville, MD (2018). URL: urlhttps://dhsprogram.com/pubs/pdf/DHSG4/Recode7_DHS_10Sep2018_DHSG4.pdf.

T.~N. Croft, C.~K. Allen, B.~W. Zachary, Guide to DHS Statistics, Technical Report, ICF, Rockville, MD (2023). URL: urlhttps://dhsprogram.com/data/guide-to-dhs-statistics/.

National Institute of Population Research and Training (NIPORT) and ICF, Bangladesh Demographic and Health Survey 2022: Final Report, Dhaka, Bangladesh and Rockville, MD (2024).

A.~R. Omran, The Epidemiologic Transition: A Theory of the Epidemiology of Population Change, The Milbank Quarterly 83~(4) (2005) 731–757. doi10.1111/j.1468-0009.2005.00398.x.

J.~R. Khan, N.~Awan, A comprehensive analysis on child mortality and its determinants in Bangladesh using frailty models, Archives of Public Health 75~(1) (2017) 58. doi10.1186/s13690-017-0224-6.

M.~A. Khan, N.~Khan, O.~Rahman, G.~Mustagir, K.~Hossain, R.~Islam, Trends and projections of under-5 mortality in Bangladesh including the effects of maternal high-risk fertility behaviours and use of healthcare services, PLOS ONE 16~(2) (2021) e0246210. doi10.1371/journal.pone.0246210.

T.~Mazumder, I.~Mohanty, D.~Ahmad, T.~Niyonsenga, An analysis of cause-specific under-5 mortality in Bangladesh using the demographic and health survey 2011 and 2017–2018, BMC Pediatrics 24~(1) (2024) 572. doi10.1186/s12887-024-04979-6.

T.~Mazumder, I.~Mohanty, D.~Ahmad, T.~Niyonsenga, An explanation of the stagnant under-5 mortality rate in Bangladesh using multilevel, multivariable analysis of three Demographic and Health Surveys, Scientific Reports 14~(1) (2024) 19823. doi10.1038/s41598-024-69924-0.

A.~M.~R. Chowdhury, A.~Bhuiya, M.~E. Chowdhury, S.~Rasheed, Z.~Hussain, L.~C. Chen, The Bangladesh paradox: exceptional health achievement despite economic poverty, The Lancet 382~(9906) (2013) 1734–1745. doi10.1016/S0140-6736(13)62148-0.

S.~E. Arifeen, A.~Christou, L.~Reichenbach, F.~A. Osman, K.~Azad, K.~S. Islam, F.~Ahmed, H.~Perry, D.~H. Peters, Community-based approaches and partnerships: innovations in health-service delivery in Bangladesh, The Lancet 382~(9909) (2013) 2012–2026. doi10.1016/S0140-6736(13)62149-2.

S.~Naznin, M.~J. Uddin, A.~Kabir, Identifying determinants of under-5 mortality in Bangladesh: A machine learning approach with BDHS 2022 data, PLOS ONE 20~(6) (2025) e0324825. doi10.1371/journal.pone.0324825.

M.~M.~M. Fahim, M.~M. Huq, M.~Sabiruzzaman, M.~R. Karim, Temporal Validation Changes the Apparent Public-Health Utility of Under-Five Mortality Prediction in Bangladesh: A Four-Round DHS Machine-Learning Study, arXiv preprint arXiv:2602.03957 (2026).

M.~A. Haque, M.~A. Kabir, Machine Learning-Based Prediction of Childhood Stunting in Bangladesh: Fairness and Temporal Robustness Assessment, arXiv preprint arXiv:2609.24386 (2026).

S.~Oduse, T.~Zewotir, D.~North, Application of machine learning methods for predicting under-five mortality: analysis of Nigerian demographic health survey 2018 dataset, BMC Medical Informatics and Decision Making 24~(1) (2024) 86. doi10.1186/s12911-024-02476-5.

S.~Mabula, R.~Too, G.~Kerich, Comparative analysis of machine learning algorithms for predicting under-five mortality: evidence from Tanzania Demographic and Health Survey, Machine Learning Research 10~(2) (2025) 110–123. doi10.11648/j.mlr.20251002.12.

F.~H. Bitew, S.~H. Nyarko, L.~Potter, C.~S. Sparks, Machine learning approach for predicting under-five mortality determinants in Ethiopia: evidence from the 2016 Ethiopian Demographic and Health Survey, Genus 76~(1) (2020) 37. doi10.1186/s41118-020-00106-2.

R.~K. Saroj, P.~K. Yadav, R.~Singh, O.~N. Chilyabanyama, Machine learning algorithms for understanding the determinants of under-five mortality, BioData Mining 15~(1) (2022) 20. doi10.1186/s13040-022-00308-8.

A.~A. Asnake, et~al., Prediction of under-five mortality using supervised machine learning algorithms in the 23 sub-Saharan African countries, Scientific Reports 16 (2026) 9131. doi10.1038/s41598-026-40401-0.

T.~Chen, C.~Guestrin, XGBoost: A Scalable Tree Boosting System, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794 (2016). doi10.1145/2939672.2939785.

G.~Ke, Q.~Meng, T.~Finley, T.~Wang, W.~Chen, W.~Ma, Q.~Ye, T.-Y. Liu, LightGBM: A Highly Efficient Gradient Boosting Decision Tree, in: Advances in Neural Information Processing Systems 30 (2017) 3146–3154.

L.~Breiman, Random Forests, Machine Learning 45~(1) (2001) 5–32. doi10.1023/A:1010933404324.

R.~Tibshirani, Regression Shrinkage and Selection via the Lasso, Journal of the Royal Statistical Society: Series B (Methodological) 58~(1) (1996) 267–288. doi10.1111/j.2517-6161.1996.tb02080.x.

N.~Meinshausen, P.~B"uhlmann, Stability selection, Journal of the Royal Statistical Society: Series B (Statistical Methodology) 72~(4) (2010) 417–473. doi10.1111/j.1467-9868.2010.00740.x.

M.~B. Kursa, W.~R. Rudnicki, Feature Selection with the Boruta Package, Journal of Statistical Software 36~(11) (2010) 1–13. doi10.18637/jss.v036.i11.

J.~C. Platt, Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods, in: Advances in Large Margin Classifiers, MIT Press, pp. 61–74 (1999).

A.~J. Vickers, E.~B. Elkin, Decision Curve Analysis: A Novel Method for Evaluating Prediction Models, Medical Decision Making 26~(6) (2006) 565–574. doi10.1177/0272989X06295361.

B.~Van~Calster, D.~J. McLernon, M.~van Smeden, L.~Wynants, E.~W. Steyerberg, Calibration: the Achilles heel of predictive analytics, BMC Medicine 17~(1) (2019) 230. doi10.1186/s12916-019-1466-7.

G.~S. Collins, K.~G.~M. Moons, P.~Dhiman, R.~D. Riley, A.~L. Beam, B.~Van~Calster, M.~Ghassemi, X.~Liu, J.~B. Reitsma, M.~van Smeden, et~al., TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods, BMJ 385 (2024) e078378. doi10.1136/bmj-2023-078378.

R.~F. Wolff, K.~G.~M. Moons, R.~D. Riley, P.~F. Whiting, M.~Westwood, G.~S. Collins, J.~B. Reitsma, J.~Kleijnen, S.~Mallett, PROBAST: A Tool to Assess the Risk of Bias and Applicability of Prediction Model Studies, Annals of Internal Medicine 170~(1) (2019) 51–58. doi10.7326/M18-1376.

T.~Lumley, Analysis of Complex Survey Samples, Journal of Statistical Software 9~(1) (2004) 1–19. doi10.18637/jss.v009.i08.

J.~Wieczorek, C.~Guerin, T.~McMahon, K-fold cross-validation for complex sample surveys, Stat 11~(1) (2022) e454. doi10.1002/sta4.454.

Y.~Oh, H.~W. Zheng, J.~Feng, A.~A.~T. Bui, Survey-aware machine learning: a guideline for valid population health inference based on scoping review, in: Proceedings of the 7th Conference on Health, Inference, and Learning, volume 333 of Proceedings of Machine Learning Research, PMLR, 2026, pp. 198–220.

H.~Shimodaira, Improving predictive inference under covariate shift by weighting the log-likelihood function, Journal of Statistical Planning and Inference 90~(2) (2000) 227–244. doi10.1016/S0378-3758(00)00115-4.

M.~Sugiyama, M.~Kawanabe, Machine Learning in Non-Stationary Environments: Introduction to Covariate Shift Adaptation, MIT Press (2012).

G.~Casella, An introduction to empirical Bayes data analysis, The American Statistician 39~(2) (1985) 83–87. doi10.1080/00031305.1985.10479401.

B.~Efron, Large-Scale Inference: Empirical Bayes Methods for Estimation, Testing, and Prediction, Cambridge University Press (2012).

S.~M. Lundberg, S.-I. Lee, A Unified Approach to Interpreting Model Predictions, in: Advances in Neural Information Processing Systems 30 (2017) 4765–4774.

D.~A. Vyas, L.~G. Eisenstein, D.~S. Jones, Hidden in Plain Sight — Reconsidering the Use of Race Correction in Clinical Algorithms, New England Journal of Medicine 383~(9) (2020) 874–882. doi10.1056/NEJMp2008447.

Z.~Obermeyer, B.~Powers, C.~Vogeli, S.~Mullainathan, Dissecting racial bias in an algorithm used to manage the health of populations, Science 366~(6464) (2019) 447–453. doi10.1126/science.aax2342.

N.~V. Chawla, K.~W. Bowyer, L.~O. Hall, W.~P. Kegelmeyer, SMOTE: Synthetic Minority Over-sampling Technique, Journal of Artificial Intelligence Research 16 (2002) 321–357. doi10.1613/jair.953.

N.~MacNell, L.~Feinstein, J.~Wilkerson, et~al., Implementing machine learning methods with complex survey data: lessons learned on the impacts of accounting sampling weights in gradient boosting, PLOS ONE 18~(3) (2023) e0280387. doi10.1371/journal.pone.0280387.

T.~Saito, M.~Rehmsmeier, The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets, PLOS ONE 10~(3) (2015) e0118432. doi10.1371/journal.pone.0118432.

References

[1] World Health Organization, Levels and trends in child mortality: Report 2023, Technical Report, Geneva: World Health Organization (2023). URL: urlhttps://www.who.int/publications/i/item/9789240081587.

[2] United Nations Inter-agency Group for Child Mortality Estimation, Levels & trends in child mortality: Report 2023, UNICEF, New York (2023). URL: urlhttps://data.unicef.org/resources/levels-and-trends-in-child-mortality/.

[3] R.~E. Black, S.~Cousens, H.~L. Johnson, J.~E. Lawn, I.~Rudan, D.~G. Bassani, P.~Jha, H.~Campbell, C.~F. Walker, R.~Cibulskis, T.~Eisele, L.~Liu, C.~Mathers, Global, regional, and national causes of child mortality in 2008: a systematic analysis, The Lancet 375~(9730) (2010) 1969–1987. doi10.1016/S0140-6736(10)60549-1.

[4] D.~You, L.~Hug, S.~Ejdemyr, P.~Idele, D.~Hogan, C.~Mathers, P.~Gerland, J.~R. New, L.~Alkema, Global, regional, and national levels and trends in under-5 mortality between 1990 and 2015, with scenario-based projections to 2030: a systematic analysis by the UN Inter-agency Group for Child Mortality Estimation, The Lancet 386~(10010) (2015) 2275–2286. doi10.1016/S0140-6736(15)00120-8.

[5] The DHS Program, DHS-VII Recode Manual, Technical Report, ICF, Rockville, MD (2018). URL: urlhttps://dhsprogram.com/pubs/pdf/DHSG4/Recode7_DHS_10Sep2018_DHSG4.pdf.

[6] T.~N. Croft, C.~K. Allen, B.~W. Zachary, Guide to DHS Statistics, Technical Report, ICF, Rockville, MD (2023). URL: urlhttps://dhsprogram.com/data/guide-to-dhs-statistics/.

[7] National Institute of Population Research and Training (NIPORT) and ICF, Bangladesh Demographic and Health Survey 2022: Final Report, Dhaka, Bangladesh and Rockville, MD (2024).

[8] A.~R. Omran, The Epidemiologic Transition: A Theory of the Epidemiology of Population Change, The Milbank Quarterly 83~(4) (2005) 731–757. doi10.1111/j.1468-0009.2005.00398.x.

[9] J.~R. Khan, N.~Awan, A comprehensive analysis on child mortality and its determinants in Bangladesh using frailty models, Archives of Public Health 75~(1) (2017) 58. doi10.1186/s13690-017-0224-6.

[10] M.~A. Khan, N.~Khan, O.~Rahman, G.~Mustagir, K.~Hossain, R.~Islam, Trends and projections of under-5 mortality in Bangladesh including the effects of maternal high-risk fertility behaviours and use of healthcare services, PLOS ONE 16~(2) (2021) e0246210. doi10.1371/journal.pone.0246210.

[11] T.~Mazumder, I.~Mohanty, D.~Ahmad, T.~Niyonsenga, An analysis of cause-specific under-5 mortality in Bangladesh using the demographic and health survey 2011 and 2017–2018, BMC Pediatrics 24~(1) (2024) 572. doi10.1186/s12887-024-04979-6.

[12] T.~Mazumder, I.~Mohanty, D.~Ahmad, T.~Niyonsenga, An explanation of the stagnant under-5 mortality rate in Bangladesh using multilevel, multivariable analysis of three Demographic and Health Surveys, Scientific Reports 14~(1) (2024) 19823. doi10.1038/s41598-024-69924-0.

[13] A.~M.~R. Chowdhury, A.~Bhuiya, M.~E. Chowdhury, S.~Rasheed, Z.~Hussain, L.~C. Chen, The Bangladesh paradox: exceptional health achievement despite economic poverty, The Lancet 382~(9906) (2013) 1734–1745. doi10.1016/S0140-6736(13)62148-0.

[14] S.~E. Arifeen, A.~Christou, L.~Reichenbach, F.~A. Osman, K.~Azad, K.~S. Islam, F.~Ahmed, H.~Perry, D.~H. Peters, Community-based approaches and partnerships: innovations in health-service delivery in Bangladesh, The Lancet 382~(9909) (2013) 2012–2026. doi10.1016/S0140-6736(13)62149-2.

[15] S.~Naznin, M.~J. Uddin, A.~Kabir, Identifying determinants of under-5 mortality in Bangladesh: A machine learning approach with BDHS 2022 data, PLOS ONE 20~(6) (2025) e0324825. doi10.1371/journal.pone.0324825.

[16] M.~M.~M. Fahim, M.~M. Huq, M.~Sabiruzzaman, M.~R. Karim, Temporal Validation Changes the Apparent Public-Health Utility of Under-Five Mortality Prediction in Bangladesh: A Four-Round DHS Machine-Learning Study, arXiv preprint arXiv:2602.03957 (2026).

[17] M.~A. Haque, M.~A. Kabir, Machine Learning-Based Prediction of Childhood Stunting in Bangladesh: Fairness and Temporal Robustness Assessment, arXiv preprint arXiv:2609.24386 (2026).

[18] S.~Oduse, T.~Zewotir, D.~North, Application of machine learning methods for predicting under-five mortality: analysis of Nigerian demographic health survey 2018 dataset, BMC Medical Informatics and Decision Making 24~(1) (2024) 86. doi10.1186/s12911-024-02476-5.

[19] S.~Mabula, R.~Too, G.~Kerich, Comparative analysis of machine learning algorithms for predicting under-five mortality: evidence from Tanzania Demographic and Health Survey, Machine Learning Research 10~(2) (2025) 110–123. doi10.11648/j.mlr.20251002.12.

[20] F.~H. Bitew, S.~H. Nyarko, L.~Potter, C.~S. Sparks, Machine learning approach for predicting under-five mortality determinants in Ethiopia: evidence from the 2016 Ethiopian Demographic and Health Survey, Genus 76~(1) (2020) 37. doi10.1186/s41118-020-00106-2.

[21] R.~K. Saroj, P.~K. Yadav, R.~Singh, O.~N. Chilyabanyama, Machine learning algorithms for understanding the determinants of under-five mortality, BioData Mining 15~(1) (2022) 20. doi10.1186/s13040-022-00308-8.

[22] A.~A. Asnake, et~al., Prediction of under-five mortality using supervised machine learning algorithms in the 23 sub-Saharan African countries, Scientific Reports 16 (2026) 9131. doi10.1038/s41598-026-40401-0.

[23] T.~Chen, C.~Guestrin, XGBoost: A Scalable Tree Boosting System, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794 (2016). doi10.1145/2939672.2939785.

[24] G.~Ke, Q.~Meng, T.~Finley, T.~Wang, W.~Chen, W.~Ma, Q.~Ye, T.-Y. Liu, LightGBM: A Highly Efficient Gradient Boosting Decision Tree, in: Advances in Neural Information Processing Systems 30 (2017) 3146–3154.

[25] L.~Breiman, Random Forests, Machine Learning 45~(1) (2001) 5–32. doi10.1023/A:1010933404324.

[26] R.~Tibshirani, Regression Shrinkage and Selection via the Lasso, Journal of the Royal Statistical Society: Series B (Methodological) 58~(1) (1996) 267–288. doi10.1111/j.2517-6161.1996.tb02080.x.

[27] N.~Meinshausen, P.~B"uhlmann, Stability selection, Journal of the Royal Statistical Society: Series B (Statistical Methodology) 72~(4) (2010) 417–473. doi10.1111/j.1467-9868.2010.00740.x.

[28] M.~B. Kursa, W.~R. Rudnicki, Feature Selection with the Boruta Package, Journal of Statistical Software 36~(11) (2010) 1–13. doi10.18637/jss.v036.i11.

[29] J.~C. Platt, Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods, in: Advances in Large Margin Classifiers, MIT Press, pp. 61–74 (1999).

[30] A.~J. Vickers, E.~B. Elkin, Decision Curve Analysis: A Novel Method for Evaluating Prediction Models, Medical Decision Making 26~(6) (2006) 565–574. doi10.1177/0272989X06295361.

[31] B.~Van~Calster, D.~J. McLernon, M.~van Smeden, L.~Wynants, E.~W. Steyerberg, Calibration: the Achilles heel of predictive analytics, BMC Medicine 17~(1) (2019) 230. doi10.1186/s12916-019-1466-7.

[32] G.~S. Collins, K.~G.~M. Moons, P.~Dhiman, R.~D. Riley, A.~L. Beam, B.~Van~Calster, M.~Ghassemi, X.~Liu, J.~B. Reitsma, M.~van Smeden, et~al., TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods, BMJ 385 (2024) e078378. doi10.1136/bmj-2023-078378.

[33] R.~F. Wolff, K.~G.~M. Moons, R.~D. Riley, P.~F. Whiting, M.~Westwood, G.~S. Collins, J.~B. Reitsma, J.~Kleijnen, S.~Mallett, PROBAST: A Tool to Assess the Risk of Bias and Applicability of Prediction Model Studies, Annals of Internal Medicine 170~(1) (2019) 51–58. doi10.7326/M18-1376.

[34] T.~Lumley, Analysis of Complex Survey Samples, Journal of Statistical Software 9~(1) (2004) 1–19. doi10.18637/jss.v009.i08.

[35] J.~Wieczorek, C.~Guerin, T.~McMahon, K-fold cross-validation for complex sample surveys, Stat 11~(1) (2022) e454. doi10.1002/sta4.454.

[36] Y.~Oh, H.~W. Zheng, J.~Feng, A.~A.~T. Bui, Survey-aware machine learning: a guideline for valid population health inference based on scoping review, in: Proceedings of the 7th Conference on Health, Inference, and Learning, volume 333 of Proceedings of Machine Learning Research, PMLR, 2026, pp. 198–220.

[37] H.~Shimodaira, Improving predictive inference under covariate shift by weighting the log-likelihood function, Journal of Statistical Planning and Inference 90~(2) (2000) 227–244. doi10.1016/S0378-3758(00)00115-4.

[38] M.~Sugiyama, M.~Kawanabe, Machine Learning in Non-Stationary Environments: Introduction to Covariate Shift Adaptation, MIT Press (2012).

[39] G.~Casella, An introduction to empirical Bayes data analysis, The American Statistician 39~(2) (1985) 83–87. doi10.1080/00031305.1985.10479401.

[40] B.~Efron, Large-Scale Inference: Empirical Bayes Methods for Estimation, Testing, and Prediction, Cambridge University Press (2012).

[41] S.~M. Lundberg, S.-I. Lee, A Unified Approach to Interpreting Model Predictions, in: Advances in Neural Information Processing Systems 30 (2017) 4765–4774.

[42] D.~A. Vyas, L.~G. Eisenstein, D.~S. Jones, Hidden in Plain Sight — Reconsidering the Use of Race Correction in Clinical Algorithms, New England Journal of Medicine 383~(9) (2020) 874–882. doi10.1056/NEJMp2008447.

[43] Z.~Obermeyer, B.~Powers, C.~Vogeli, S.~Mullainathan, Dissecting racial bias in an algorithm used to manage the health of populations, Science 366~(6464) (2019) 447–453. doi10.1126/science.aax2342.

[44] N.~V. Chawla, K.~W. Bowyer, L.~O. Hall, W.~P. Kegelmeyer, SMOTE: Synthetic Minority Over-sampling Technique, Journal of Artificial Intelligence Research 16 (2002) 321–357. doi10.1613/jair.953.

[45] N.~MacNell, L.~Feinstein, J.~Wilkerson, et~al., Implementing machine learning methods with complex survey data: lessons learned on the impacts of accounting sampling weights in gradient boosting, PLOS ONE 18~(3) (2023) e0280387. doi10.1371/journal.pone.0280387.

[46] T.~Saito, M.~Rehmsmeier, The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets, PLOS ONE 10~(3) (2015) e0118432. doi10.1371/journal.pone.0118432.